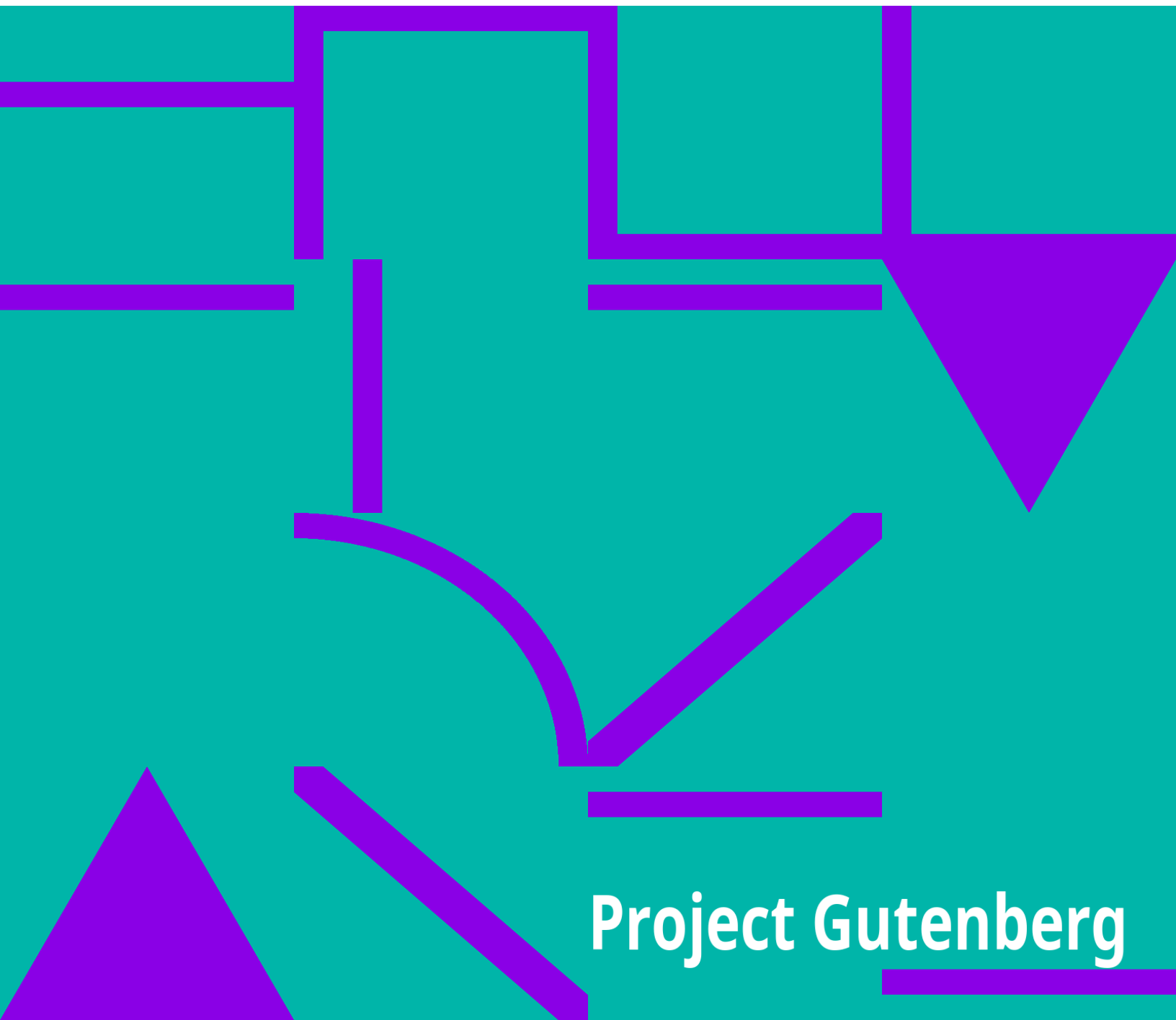


Colleges in America

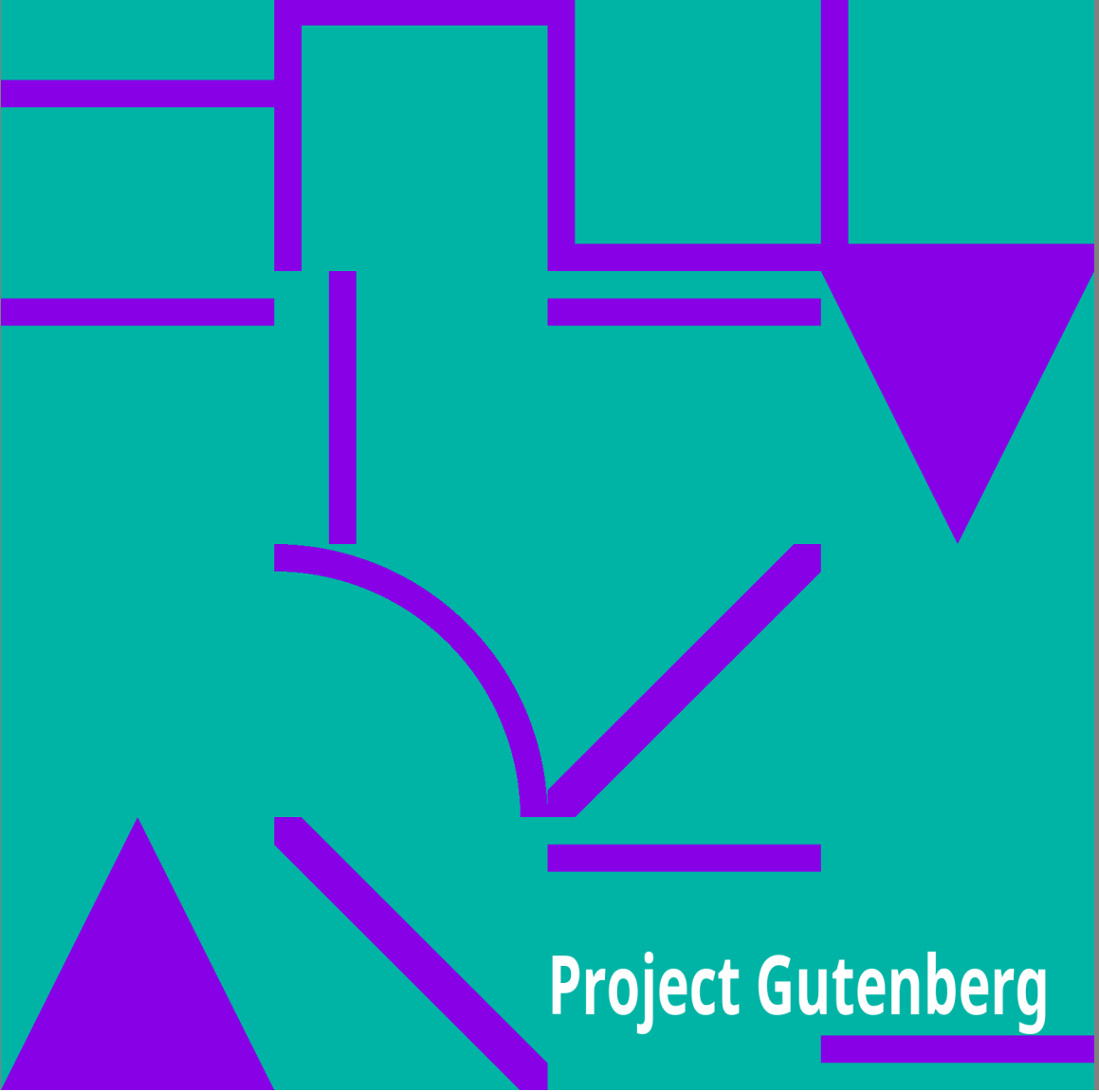
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Colleges in America

John Marshall Barker



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AMERICA ***

COLLEGES IN AMERICA.

BY

JOHN MARSHALL BARKER, PH. D.

WITH AN INTRODUCTION BY

REV. SYLVESTER F. SCOVEL, LL. D.,

PRESIDENT OF THE UNIVERSITY OF WOOSTER.



THE CLEVELAND PRINTING & PUBLISHING CO.,
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TO ONE OF THE
GREATEST LIVING SCHOLARS AND EDUCATORS,
REV. WILLIAM F. WARREN, LL. D.,
PRESIDENT OF BOSTON UNIVERSITY.

NOTE.

The author of this volume aims to give the reader a brief survey of the growth, functions, and work of the American Colleges. It has been a pleasure to visit many of the colleges and gather facts, receive impressions and carry away many pleasant recollections regarding them.

The following authorities have been helpful in the preparation of the work: "A History of Education," by F. V. N. Painter; "The Rise and Early Constitution of Universities," by S. S. Laurie; "Education in the United States," by Richard G. Boone; "Essays on Educational Reformers," by Robert H. Quick; "Education," by Herbert Spencer; "Universities in Germany," by J. M. Hart; Huxley's "Technical Education;" Froude's "Essay on Education,;" "The American College and the American Public," by President Noah Porter; "Prayer for Colleges," by Professor W. S. Tyler; "American Colleges: their Life and Work," and "Within College Walls," by President Chas. F. Thwing; "Universities on the Continent," and "Culture and Anarchy," by Matthew Arnold; "Educational Essays," by Bishop Edward Thomson; "Christianity in the United States," by Daniel Dorchester; "College Life," by Stephen Olin; "The Intellectual Life," by P. G. Hamerton; "Essays on a Liberal Education," by F. W. Farrar; "History of Higher Education" in the several States, prepared by the Bureau of Education; "Reports of the Commissioner of Education for 1890-'91;" and the periodical literature bearing on the subject.

CONTENTS.

I.	The Rise of Universities in the Old World,	<u>13</u>
II.	The Planting of Colleges in the New World,	<u>36</u>
III.	Characteristics of the American College,	<u>69</u>
IV.	The Functions of the American College,	<u>104</u>
	<i>a.</i> A Symmetrical Development.	
	<i>b.</i> The Advancement of Knowledge.	
	<i>c.</i> Preparation for Service.	
V.	Student Life in College,	<u>156</u>
VI.	The Personal Factors in a College Education,	<u>178</u>
VII.	The Practical Value of an Education,	<u>196</u>
VIII.	Our Indebtedness to Colleges,	<u>229</u>

INTRODUCTION.

I cannot be unwilling to avail myself of any opportunity to turn the attention of the Christian public to the Christian College. It is a noble public and an equally noble object. I can conceive of no worthier or more Christian thing than the caretaking of one generation that the next one which must necessarily lie so long under its influence and for which it is therefore so thoroughly responsible, should receive a Christian education.

To put Christ at the center and make Him felt to the circumference (as Bungener said in speaking of Calvin's school policy), is exceedingly difficult. But it is exceedingly important. It is, indeed, vital and pivotal.

The dangers about it are great and ever greater. They come from the general worldliness of all things and everybody in this age of unprecedentedly rapid and splendid material development. They are increased by the growth of speculative infidelity whether of the philosophical or scientific phase. They spring out of everything which lowers the Bible from that supreme and sovereign consideration by which alone it can hold the place in education which the Old Testament economy gave it, and which all the books of all the other book-religions of the world most unquestioningly possess. They are born of all that false theorizing about the limits of government and the liberty of conscience which issues in the demands for utter secularization of every institution of the State, while at the same time the necessities of popular government are demonstrating that education must be by the State. They are intensified by the divided opinion of the church universal, of which the Catholic and Greek sections hold that education must be religious and under the care of the Church; while the State-Church Protestant section holds that it may be religious under certain conditions, and the extreme secularistic protestant wing holds that it cannot be religious because conducted by the State, and a rather diminishing protestant section in free-church nations holds that the higher education should be Christian, while the secondary and primary may safely be left to the secular State.

These dangers are not only imminent but actual. The whole effort to support a Christian education in the public schools is sometimes called a "bootless wrangle." One section is thrown over towards secularism, pure and simple, in recoiling from Church-education exclusive and reactionary. The leading of the little child, the favorite indication of the millennium's arrival, is frustrated amid the clamor of the free thinkers and the uncertainty of the Church and the necessities of the State. We are slowly but surely, if we go on in this way, taking our children out of Christ's arms and our youth from beside His footsteps. And that is at once the most fearful sin against Him, and the most terrible injustice to them, we could possibly commit. Who can do anything to stay this destructive tendency? "God bless him," I would say in Livingstone's spirit, "whoever he may be," that will help to heal this open wound of the world.

I think Mr. Barker's little book will help. It supplies much information carefully collected from scattered sources, given in brief and explicit statements. Its range of themes is wide and upon them all some standard thoughts are given. It is addressed to all readers and should find them among parents (whom it should make patrons), among those who have hearts to pray and those who have hands to help. It will prove to be of rare interest to all whose duty it is to teach, and it has much wise counsel for those who are to study.

The treatment of the function of the College for the cultivation of the moral and spiritual nature (Chapter IV) deserves special attention. Its declarations are firm, its ideals high and its selected opinions apt and forcible. It ought to end the reign of any institution in which religion is not put at the center and kept as efficient as human instrumentalities can make it. The demand for professors of pronounced Christian character and convictions is timely and is fearlessly made.

The discussion of the currents and counter-currents of influences in college life cannot but be useful, with a possibly increased emphasis against the secret societies and a caution against organizations of undergraduates for active partisan work in politics. The time for these fruits is "not yet."

Admirably the author shows that we have the best College material in the world and that it behaves itself best. And there can be no lack of agreement

as to the arousing arguments and the closing chapters concerning the usefulness of colleges to the individual and the community. May it serve to kindle and to extend when kindled the wholesome enthusiasm its respected author manifests both by word and work.

SYLVESTER F. SCOVEL.

The University of Wooster,
July 9, 1894.

COLLEGES IN AMERICA.

I.

THE RISE OF UNIVERSITIES IN THE OLD WORLD.

The American college system is deeply rooted in the past. It will be better understood if we trace briefly its historic connection with the ancient and European seats of learning. Higher education has been promoted among all great nations. Flourishing colleges were founded among ancient people. In the kingdoms of Judah and Israel, schools of the Prophets were located at Bethel, Gibeah, Gilgal, Jericho and Naioth. The Academy of Athens, the Museum of Alexandria, the Athenæum of Rome were once centers of intellectual activity and spread their influence over the civilized world.

The Greek race especially commands our attention for its activity in matters relating to higher education. The Academy of Plato flourished for nine hundred years. The schools of Athens are noted for their great and permanent influence in awakening thought and shedding the light of their teaching among the nations of the world. "So charged," says Cardinal Newman, "is the moral atmosphere of the East with Greek civilization, that down to this day those tribes are said to show to most advantage which can claim relation of place and kin with Greek colonies established two thousand years ago." The influences of the scholastic halls of Plato and Aristotle span the centuries with their light and power.

Here truths were taught that have found universal acceptance. Down to the second century, Athens was a favorite resort for students. The college at Alexandria, where so many of the Fathers of the Church were educated, was founded and carefully organized by Ptolemy two centuries before the Christian era. For six hundred years it exerted a great influence on the youth

who gathered from all parts of the civilized world to receive instruction from its eminent professors.

Roman colleges likewise exerted a wholesome influence in their day. They began during the life-time of Quintilian, in the second century, and it continued to be the deliberate policy of Augustus, Vespasian and Hadrian to multiply and extend the influence of endowed schools in Rome and provincial towns. Their object, says Merivale, was to "restore the tone of society and infuse into the national mind healthier sentiments." These Romano-Hellenic schools were so tenacious of life that they continued to flourish down to the fifth century. Owing to the decline of personal morality and the low conceptions of the ends of human life, and other general influences which led to the downfall of the empire, these schools finally degenerated and could no longer survive.

"Some great new spiritual force," says Professor Laurie, "was needed to reform society and the education of the young. That force was at hand in Christianity; and if it very early assumed a negative, if not a prohibitory, attitude to the old learning, it may be conceded that this was an inevitable step in the development of a new ethical idea."

The Christian system of education gradually superseded the pagan system. Christianity fortified the sense of personality and introduced the idea of a broader and deeper sentiment of human brotherhood, which helped to diffuse the spirit of education among the people and awaken in the human mind a sense of its native dignity and power.

There were in the first century such men as Clemens, Ignatius and Polycarp, who employed their talent to build up Christianity and encourage the education of the people. In the second century, "the number of the learned men increased considerably, the majority of whom were philosophers attached to the elective system." It was at the close of this century (181 A. D.) that the first Christian catechetical school was established at Alexandria, in accord with Christian requirements. Such schools soon became numerous and efficient, and were under the superintendence of the Bishops. The priests, as well as the laity, were educated in them. At the end of the fourth century they had entirely superseded the schools of the *grammaticus*, when ancient culture became practically extinct.

The monastic schools arose in the fifth century to supplant the Romano-Hellenic schools. Chief among the founders in the West was Benedict, who in 428 A. D. founded a monastery on Monte Cassino, near Naples. "He had educational as well as religious aims from the first, and it is to the monks of this rapidly extending order, or to the influence which their 'rule' exercised on other conventual orders, such as the Columban, that we owe the diffusion of schools in the early part of the Middle Ages and the preservation of ancient learning. The Benedictine monks not only taught in their own monasteries, but were everywhere in demand as heads of Episcopal or Cathedral schools."[\[A\]](#)

[\[A\]](#) Laurie.

The monastic schools multiplied rapidly throughout Europe and took the lead in education and gained more influence than the episcopal schools. These schools, sheltered by the church, existed from the fourth to the twelfth century for the benefit of the ecclesiastical body. The majority of them did not admit lay instruction until the middle of the ninth century. Education during this period, with few exceptional centers, was crude and unenlightened. The power of the mediæval machinery was such that these schools gave to the clergy only the mere rudiments of learning. The conception of education at first did not embrace the culture of the whole man. It was commonly thought that the religious life opposed the life of the world, and that the temporal life should be one of abnegation and asceticism. It was the belief that human reason could not be trusted to have independent activity, and so dogma was substituted for its free movement. The mind was cribbed and confined by rules, for fear that speculations in philosophy and free investigations would disturb and rationalize theology. Thought was so fettered that philosophy, literature and science were almost forgotten. Everything was done to subserve the faith and suppress heresy. The Latin and Greek classics were denounced as the offspring of the pagan world. It required several centuries for the Christian world to conceive that there was no antagonism between reason and authority, and between Greek and Roman culture and the Christian religion. These schools, however, did a valuable service to the cause of education by transcribing manuscripts and becoming repositories of ancient learning.

The intellectual chaos began to end about the tenth century. The re-establishment of civilization and the revival of learning was still more manifest during the eleventh century, and soon university life became possible. The time was evidently ripe for Europe to awake from its intellectual sleep and begin a new educational development. The general causes which contributed to give fresh impulse to higher education at this time were the growing tendency to organization, the Saracen influence and the desire for higher learning in the more important centers. "The universities were founded," says Professor Laurie, "by a concurrence of able men who had something they wished to teach, and of youth who desired to learn. * * * It was the eternal need of the human spirit in its relation to the unseen that originated the University of Paris. We may say then that it was the improvement of the professions of medicine, law and theology which led to the inception and organization of the first great schools."

The people felt the need of providing and obtaining instruction beyond the monastic and episcopal schools. By the natural development of these, a number of high-grade schools were established which afterwards gave rise to the universities. They came into existence without charter from either ecclesiastical or civil power, and were not controlled or directed by either. The importance of these institutions was soon discovered by both Pope and Emperor, who cultivated friendly relations with these free, voluntary and self-supporting centers of learning and gave them special privileges and encouragement.

Among the first European schools was that of Salerno, in Italy, which was known as a school of medicine as early as the ninth century. The University of Bologna arose at the close of the twelfth century. In 1211 the University of Paris became a legal corporation. Oxford began as a secondary school, and passed to the rank of a university in 1140, and Cambridge was established in the year 1200. Professor Laurie says that "in the course of the twelfth and thirteenth centuries there grew up in Europe ten universities; while in the fourteenth century we find eighteen added; and in the fifteenth century twenty-nine arose, including St. Andrew's (1411), Glasgow (1454), Aberdeen (1477). The great intellectual activity of the fourteenth century, which led to the rise of so many universities, coincides with the first revival

of letters, or rather was one manifestation of the revival." The main center of this great intellectual movement was the University of Paris, the mother of universities, which gained pre-eminence in the great studies of theology and philosophy. It was chartered by Philip Augustus in the thirteenth century, and was fostered by France, Picardy, Normandy and England. These united and organized the Faculty of Arts, which became its chief glory. It taught the three arts, Latin grammar, rhetoric and dialectics, known as the *trivium*. The *quadrivium*, embracing arithmetic, geometry, astronomy and music, was likewise taught. The Faculty of Theology was created in 1257, that of Law in 1271, and that of Medicine in 1274.

Matthew Arnold says that "the University of Paris was the main center of mediæval science, and the authoritative school of mediæval teaching. It received names expressing the most enthusiastic devotion, the *Fountain of Knowledge*, the *Tree of Life*, the *Candlestick of the House of the Lord*. * * * Here came Roger Bacon, Saint Thomas Aquinas and Dante; here studied the founder of the first university of the empire, Charles the Fourth, Emperor of Germany and King of Bohemia, founder of the University of Prague."

The intellectual lead which belonged to France in the twelfth and thirteenth centuries passed to Italy in the fourteenth century. Some of the universities in Italy ranked among the best in Europe. They were chiefly distinguished for their studies in law and medicine. In the early part of the thirteenth century, the University of Bologna was famous throughout the world, having at one time 12,000 students from all parts of Europe. These universities continued to exert a powerful influence until Catholicism triumphed over the abortive attempts at religious reform, and there settled down over the brilliant Italy of the Renaissance an unprogressive and anti-intellectual influence from which she has never fully recovered.

"The importance of the university in the thirteenth and fourteenth centuries," says Matthew Arnold, "was extraordinary. Men's minds were possessed with a wonderful zeal for knowledge, or what was then thought knowledge, and the University of Paris was the great fount from which this knowledge issued. The University and those depending on it, made at this time, it is said, actually a third of the population of Paris. * * * One asks oneself with interest, what was the mental food to which this vast, turbulent

multitude pressed with such inconceivable hunger. Theology was the great matter; and there is no doubt that this study was by no means always that barren and verbal trifling which an ill-informed modern contempt is fond of representing it. It is evident that around the study of theology in the mediæval University of Paris there worked a real ferment of thought, and very free thought. But the University of Paris culminated as the exclusive devotion to theological study declined, and culminated by virtue of that declension."

The great business of the universities from the twelfth to the seventeenth century was that of scholastic philosophy, which largely governed their teaching.

The scholastic philosophy was "the legitimate development of the philosophy of Aristotle and his successors, and was the only philosophy possible in its day. Nay, it was an integral essential element in human progress. It taught men to distinguish and define, and has left its impress upon the language and thought of all civilized peoples, 'in lines manifold, deep-graven and ineffaceable.' Out of it has grown our modern civilization."

The schoolmen would freely canvass the deep problems of the mind and soul, but would blindly exclude the new influences at work in society. They had to meet the opposition of the humanists, who made the study of Latin and Greek the basis of culture. The humanists were great writers and artists, who worked for more modern ideas and a newer civilization. They introduced the Renaissance, which was a literary movement that began in Italy in the fourteenth century. It was believed that vital knowledge was gained by knowing oneself, and that the best way to attain this was to study poetry, philosophy, history and all knowledge that was created by the spirit of man. Unfortunately, the knowledge of letters in Italy tended to paganize its adherents. Infidelity spread and immorality abounded in all ranks of society.

The great movement of the Renaissance secured a stronghold in Germany, where its power was extended to the established systems of instruction and utilized in the interests of a purer Christianity. Melancthon and Erasmus and all the chief reformers except Luther, were eminent humanists and friends of classical learning. They were outside the established schools, and were

the leading spirits in intellectual culture, so that the Renaissance triumphed with the Reformation. These two forces united and gave spirit and power to the humanists. The influence of the new learning in Germany was marked by comparative freedom from frivolities, skepticism and immoralities. There was a critical and enlightened study of classical literature and a reverent and rational study of the Bible. The literary treasures of antiquity were made to minister to religion. The Reformation also gave fresh impulses to all the schools and institutions of learning. The school teacher and preacher of the gospel joined hands in the common work of education.

The universities, however, under the control of the schoolmen, retrograded and decayed because they chose to remain mediæval. They refused to become the educational agencies of the times, and so failed to be at the head of a great intellectual movement. They could not be induced to assimilate the new studies and make themselves the organ of the Renaissance and the Reformation. The rapid growth of positive and experimental science, however, was fatal to scholasticism. The narrow scholastic spirit was exemplified by Cremonini, who is called the last of the schoolmen, and who was professor at Padua in 1631.

This countryman of Galileo, after the discovery of Jupiter's satellites, judging that this discovery contradicted Aristotle, would never consent to look through a telescope again. One could not have a better incident to end the career of the scholastic philosophy.

The Jesuits adopted a more liberal spirit and method. They established and controlled a large number of universities and schools, and made them the great channels of the movement of the counter-Reformation. Their educational activity gained for them a great reputation for teaching and a large patronage. In 1710, they had 612 colleges, 157 normal schools, 24 universities and 200 missions. They were inspired not so much by the value they placed on culture for its own sake, as to promote the authority of the old religion and prevent heresy.

The powerful initial impulse given to the cause of education by means of the humanists and the reformers in the fifteenth and sixteenth centuries began to flag in the seventeenth century, when the Protestant Church, like the Catholic, became cold and petrified. The universities were regarded as

appendages of the church, and classical training largely lost its hold in Europe.

The condition of contemporary institutions for superior instruction in the old world is full of promise. The importance of building up great universities is conceded by nearly all nations. In the judgment of Mr. L. D. Wishard, the Foreign Secretary of the College Y. M. C. A., there are 500,000 young men in Asia in the high-class institutions.

The government of Japan, that has lately joined the Western nations in the onward march of civilization, gives enlightened direction to higher education. There are, besides the Imperial College of Tokio, five great secondary schools located in different centers throughout the empire, which serve as feeders to the university. There are 5,000 youth in Christian colleges and schools in the kingdom. In the Christian university at Kioto there are 600 youth pursuing a college education under Christian teaching.

China has always encouraged colleges for the education of her magistrates. "The literary class consisting of the graduates, and those who attend the examinations for degrees, numbering some two and a half millions, are the rulers of China."

There is a growing tendency to universal education in India. "It is computed," says Bishop Hurst, "that in the small area of Calcutta and suburbs there are 28,000 alumni who have completed the curriculum in the five Christian colleges. There are about 2,000 who are alumni or students of the Calcutta University, and there are 1,000 youths besides who are studying up to the matriculation examinations of the university." The English language is the medium of instruction in all these institutions. It may not be wide of the mark to suppose that in all India there are not less than 40,000 natives who have graduated at some school of high grade, and that ten per cent. of the number have passed the university degrees. The number is now more probably 50,000. These men enjoy the highest respect and are the recognized leaders of native thought. Already many are, and many more are to be judges, lawyers, magistrates, professors, teachers, orators, physicians, engineers, merchants, authors and journalists of the country.

The University of Fez, in Morocco, established in the eighth century, is one of the oldest universities outside of Asia. The Mohammedan University at Cairo, in Egypt, has more than 200 instructors and 10,000 students assembled from Europe, Asia and Africa to be instructed in the Moslem faith.

If we turn to Europe, we find that the planting and enlarging of the institutions for superior instruction has the most hopeful outlook. In Great Britain and Ireland there are 11 universities with 834 professors and 18,400 students. Besides, there are the old established and excellent schools at Eaton, Harrow, Winchester and Rugby.

A new era for the classical schools of Germany began in 1783, when Baron Sedlitz, encouraged by Frederic the Great, was able to revive "the dormant sparks planted in them by the Renaissance and they awoke to a new life, which since the beginning of this century has drawn the eyes of all students of intellectual progress upon them." Germany had in 1890, 250 gymnasias and 22 universities. The latter are manned by 2,431 instructors and have 31,803 students, or one student to every 151 of the population.

France has 19,152 students in her professional and technical schools. There are fifteen institutions of higher learning in the University of France, with 180 professors and 12,695 students. These are under the control and patronage of the State. The government appropriated in 1889–90, 12,000,000 francs for university purposes. Besides, there were expended in the same year 99,000,000 francs for new buildings for the advancement of higher education. In 1890, there were 598 professional chairs in the several universities, in which were taught 17,630 students, or one student to every 217 of the population.

The Austria-Hungary Empire had in 1891 eleven universities, eight of which were in Austria, with 1,112 professors and 14,272 students. The remaining three were in Hungary and had 322 professors and 4,098 students. There were for the same year in Switzerland nine universities, with 434 professors and 2,619 students.

The Catholic Church in Italy continued for years to exert an unprogressive and anti-intellectual influence. The present government of Italy, however, is

fully awake to the importance of a university education for the people, and now maintains several universities at a large annual outlay.

This brief outline reveals the facts that all civilized nations are encouraging and maintaining schools for the higher education of the people, and suggests that a comparative study of them is both helpful and fruitful.

Many of the universities in the Old World lack the stimulus of the strong Protestant denominational influence and the marked religious character of the American colleges. They consequently fail to attain the highest results for the general good, but they are inaugurating an intellectual movement which will eventuate in a more glorious future.

II.

THE PLANTING OF COLLEGES IN THE NEW WORLD.

Our national existence came into full bloom under the light of a Christian civilization. The political, social and religious institutions were sufficiently well organized in the Old World to be advantageously introduced, with some modifications, into a young nation in the New World.

The early colonists first founded a church, then a school, and then a college. They felt that the colonial organization was incomplete without a college to inculcate such piety, virtue and intelligence as would preserve and perfect the highest social order and secure the blessings of liberty. These colleges, modelled at first after the universities of Europe, soon mapped out a pathway for themselves, and have now come to occupy a unique place in our national life.

The Pilgrim Fathers sought to establish in the New World three great principles: civil and religious liberty, and to make education their cornerstone. The scholarly impulses were so dominant at this early day that when the entire population of New England did not exceed four thousand, the people determined to establish a college, which Cotton Mather says "was

the best thing they ever thought of." It is estimated that this meager population contained as many as one hundred men who had received the training of Oxford and Cambridge. Sixty of them were from the University of Cambridge; twenty were from Oxford, and others, apparently, from the Scotch universities. The colleges they founded show traces of all these institutions. These intelligent and refined men, with breadth of culture and political foresight and public spirit, constituted the chief source of greatness in the early days of New England.

The three leading colonial colleges, Harvard, Yale, and William and Mary, were planted and permeated with the spirit of republican liberty and primitive Christianity. They began in a very modest way.

Harvard, the oldest of American colleges, was founded in the beginning of the colonial days, only eighteen years after the Pilgrim Fathers landed on Plymouth Rock, and when Boston was a village of twenty-five or thirty houses, and when only twenty-five towns had begun to be settled in the colony. In 1636, six years after the settlement of Boston, the colonial legislature voted the sum of four hundred pounds (equivalent to a tax of fifty cents to every person in the colony) towards the founding of Harvard College, with the avowed purpose of training young men for the ministry. This sum was increased in 1637 by the munificence of John Harvard, who was a graduate of Cambridge, and a finished scholar and clergyman from England. He gave eight hundred pounds and his library, consisting of three hundred volumes, towards the endowment, whereupon the college took his name. "The colony caught his spirit," says Boone. "Among the magistrates themselves, two hundred pounds was subscribed, a part in books. All did something, even the indigent; one subscribed a number of sheep; another, nine shillings' worth of cloth; one, a ten-shilling pewter flagon; others, a fruit dish, a sugar spoon, a silver-tipped jug, one great salt, one small trencher salt, etc. From such small beginnings did the institution take its start. No rank, no class of men, is unrepresented. The school was of the people." There is nothing in history to parallel the heroic spirit and boldness of these early settlers in attempting to found a college, surrounded as the people were with poverty, scanty subsistence, and savage enemies. They did not realize the wisdom of their liberality and sacrifice and its influence upon the future civilization of the Western World. Harvard College was

located at Cambridge, with a single building, on less than three acres of land. It was supported by government appropriations and private philanthropy. For years the college was financially embarrassed. The salaries were small, and for nearly one hundred years were paid out of the colonial treasury. The President received a salary of \$600. The total grants made to the college by the colony during the first century amounted to about \$8,000. The total annual income from all sources at the close of the first century of its history was but £750. Down to 1780 the total amount contributed out of the public treasury was \$68,675 and 3,793 acres of land. Individuals in England and America had likewise given \$90,412.

No one at this period would have dared to predict that Harvard College would have in 1892 an endowment of \$12,000,000 and an annual revenue of more than \$1,000,000, with seventeen departments of instruction, three hundred teachers, and three thousand students. But such has been the phenomenal growth of some of our American institutions.

Among the colonial colleges, that of William and Mary is one of the most important. As early as 1617, an attempt was made in England to raise money to found a college among the Virginia settlers. In 1619, fifteen hundred pounds were in the hands of the treasurer, and ten thousand acres of land were granted by the Virginia Company. A preparatory school was founded two years later, but owing to the Indian massacre of 340 settlers which followed, the enterprise was suspended. The effort to found a college was subsequently revived in 1660. The Virginia Assembly enacted that "for the advancement of learning, education of youth, supply of the ministry, and promotion of piety, there be land taken for a college and free school." Nothing came of this until 1688, when a subscription was taken from wealthy planters for twenty-five hundred pounds for the college. Five years later (1692) the first royal educational charter in America was granted. The college was established at Williamsburg, Virginia, and was given £2,000 and 20,000 acres of land, a tax of a penny a pound on all tobacco exported from Virginia and Maryland, and the duty on furs, skins, and liquors imported, besides other fees and privileges of the Surveyor General's office. "In its royal foundation, its generous endowment, and liberal patronage," says R. C. Boone, "it stands in sharp contrast to the early years of Harvard. This was established by the Puritans, and stood for the severest of ultra-

orthodox though dissenting Protestantism; that was founded to be and was an exponent of the most formal ceremonialism of the Church of England. The one was nursed by democracy; the other befriended by cavalier and courtier. Endowment for the one came from the purses of an infant and needy settlement; the other was drawn from the royal treasury. The one was environed and shaken for a hundred years by the schisms of a controversial people; the roots of the other were deep in the great English ecclesiastical system." This college has been called a school of statesmen. It was here that Jefferson, Randolph, Tyler, Monroe, Blair, Marshall, and other prominent statesmen received their training.

The history of Yale College is full of interest. The original design of the founders of the New Haven Colony was to establish a college. A lot was set apart for this purpose as early as 1647. A plan was proposed in 1698 to found a college, and to be placed under the general care of the churches. In 1700, sixty-three years after the founding of Harvard College, a society consisting of eleven ministers met to take the initial step. At a second meeting, in the same year, each of the trustees, numbering ten of the principal clergymen of the colony, were without money, but they brought forty volumes of books, and, placing them on a table, presented them to the body, saying in substance: "I give these books for the founding of a college in this colony." This was the humble beginning of Yale College. The colony had a population at this time of fifteen thousand people, fifty of whom were college-trained men. The outlook for this college was not very encouraging, in view of their limited means and scattered population. The work, at first, lacked system and unity. In 1718, the college was permanently located at New Haven, Connecticut, and named in honor of Elihu Yale, who was born in Boston in 1648. He received his education in England, and was afterward made Governor of Madras, and, later, Governor of the East India Company. His donation to Yale College was largely in books, and amounted to five hundred pounds. This gift was followed by that of Rev. George Berkely, who gave ninety-six acres of land in Rhode Island and one thousand volumes to the library. The college received for its support, in a century and a half, \$100,000 from the commonwealth of Connecticut. It has been supported chiefly by private means. In 1890, there were 143 instructors and 1,500 students. There is no college in America that has a more enviable

reputation for giving a thorough Christian education to the thousands of youth who have gone forth from her halls of learning.

It is a matter of record that our ancestors showed much self-denial, courage, and genius, to turn aside from the work of organizing a new social order, and the readjustment of themselves to their surroundings in a new country to provide for the higher education of the people. The founders and supporters of these colleges, as a rule, were men of high intellectual and religious character, and worked intensely and earnestly for the highest good of society. It would prove an inestimable blessing to our nation if every American citizen were inspired with the zeal of the early colonists in behalf of the cause of higher education. They, out of their poverty, poured their gifts into the treasury of the colleges in order to leave future generations a great and glorious heritage. Gratitude should prompt us to excel them in our love for the education of the present and future generations by cheerfully giving of our abundance for the same high and holy ends.

Other colleges were founded within the century. Aside from the three colonial colleges, six more were founded prior to the Revolution, and four during the war of independence. Following the Revolution was a period of expansion, and by the close of the century there were twenty-four colleges established. These colleges, scattered throughout the Union, appeared as a galaxy of stars in the literary firmament of the nation. They were founded and located as follows:

	<i>Institution.</i>	<i>State.</i>	<i>Date.</i>
1.	Harvard,	Massachusetts,	1637
2.	William and Mary,	Virginia,	1693
3.	Yale,	Connecticut,	1701
4.	Princeton,	New Jersey,	1746
5.	University of Pennsylvania,	Pennsylvania,	1749
6.	Columbia,	New York,	1754
7.	Brown,	Rhode Island,	1764
8.	Dartmouth,	New Hampshire,	1769
9.	Queen's Rutgers,	New Jersey,	1766
10.	Hamden-Sidney,	Virginia,	1776
11.	Washington and Lee,	Virginia,	1782

	<i>Institution.</i>	<i>State.</i>	<i>Date.</i>
12.	Washington University,	Maryland,	1782
13.	Dickinson,	Pennsylvania,	1783
14.	St. Johns,	Maryland,	1784
15.	Nashville,	Tennessee,	1785
16.	Georgetown,	Dist. of Columbia,	1789
17.	University of N. Carolina,	North Carolina,	1789
18.	University of Vermont,	Vermont,	1791
19.	University of E. Tennessee,	Tennessee,	1792
20.	Williams,	Massachusetts,	1793
21.	Bowdoin,	Maine,	1794
22.	Union,	New York,	1795
23.	Middlebury,	Vermont,	1795
24.	Frederick College,	Maryland,	1796

It remained for the nineteenth century to exhibit in the New World an unprecedented multiplication and expansion of institutions of higher learning.

At the opening of the century there were only twenty-four colleges in the United States. Thirty years later the number had reached forty-nine. In 1850, there were 120 colleges, manned by 1,300 teachers, with 17,000 students. There were besides 42 theological seminaries, 35 medical schools, and 12 law schools.

By 1890, the number of colleges and universities had grown to 415, having 7,918 instructors and 118,581 students. There were in the same year 117 medical schools, with 7,013 students, and 54 law schools, with 4,518 students. These facts bear witness to the determination of the American people to satisfy the needs of their higher nature, and not to rest content with material growth and the bare necessities of life.

The spirit of our early ancestors was never more manifest than in their earnest advocacy of religious liberty, and their protest against all ecclesiastical authority. The numerous settlements in different sections of the country, with their different nationalities and diverse religious opinions, tended to multiply the religious denominations and to establish churches with divergent aims and plans. These independent sects gave rise to a great

number of schools claiming to be colleges. These schools they regarded as essential and supplementary to their churches. Harvard owes its origin to non-conforming clergymen. The Episcopal Church claimed William and Mary College. The Congregationalists of Connecticut founded Yale. Princeton was founded under the auspices of a Presbyterian synod, and Brown was established by an association of Baptist Churches. One hundred and four of the first one hundred and nineteen colleges established in the United States had a distinctively Christian origin. Their founders intended that they should be, in some sense, ecclesiastical as well as religious. Notwithstanding their diversity, there was unity in their general character and design. While they maintained a denominational character, they were in nowise illiberal, and set up no religious test for entrance.

The Christian Churches have been not only pioneers of education, but their followers recognize as never before the power and efficiency of the Christian College to further the Kingdom of God on earth. Out of 415 colleges in 1890, 316 of them were under the control of some religious denomination. These were distributed in 1890 among the several denominations as follows: Methodist, 74; Presbyterian, 49; Baptist, 44; Roman Catholic, 51; Congregational, 22; Christians, 20; Lutheran, 19; United Brethren, 10; Protestant Episcopal, 6; Reformed, 6; Friends, 6; Universalist, 4; Evangelical Association, 2; German Evangelical, 1; Seventh Day Adventist, 1; New Church (Swedenborgian), 1.

The leading denominations are especially active in promoting the cause of higher education. We summarize the educational work of a few of them:

The Congregational Churches, with a membership of 525,097, had, in 1890, thirty-eight schools of distinctly college rank, with 1,034 instructors and 13,601 students. This denomination has generously endowed many of her colleges. She has been pre-eminent in her efforts to extend a liberal education to the people.

The Roman Catholic Church in the United States claimed to have, in 1894, 116 colleges, 637 academies, and 768,498 pupils in parochial schools. This church, that numbers among its adherents one-tenth of the population of this country, has one-fourth of all the colleges.

The Regular Baptists of the United States have one hundred and fifty-two chartered institutions of learning, with an endowment and property valuation of \$32,162,904. Of these, seven are theological seminaries, with 54 professors, 776 students, and \$3,701,620 of endowments and property. Thirty-five are universities and colleges open to both sexes, with 701 professors and instructors, 9,088 students, and endowment and property to the amount of \$19,171,045. Thirty-two are colleges exclusively for women, with 388 professors and instructors, 3,675 students, and endowment and property, \$4,121,906. Forty-seven are seminaries and academies, male and co-education, with 369 professors and instructors, 5,250 students, and endowment and property worth \$3,787,793. And thirty-one are institutions of learning for colored people and Indians, several of which are chartered colleges, with 279 instructors, 5,177 students, endowment and property worth \$1,380,540.

Among the church families in the United States the Presbyterians stand third, having about 1,500,000 members, 13,476 organizations, and church property valued at \$94,869,000. They have always been favorable to the higher education of ministers and people, and therefore liberal in support of the better class of schools and colleges. They now have under their immediate care 56 colleges, with an enrollment of 10,143 students. The estimated value of property owned by these institutions is \$6,780,600, and their permanent endowment funds amount to \$6,891,800. There are, besides, four colleges which are jointly owned and patronized by Presbyterians and Congregationalists. In addition there are some forty classical academies, under the care of different Synods and Presbyteries, which have over 3,000 students, and property whose net value is over \$1,000,000. Fourteen theological seminaries are scattered over the country, with more than 1,200 students. These have property and endowments amounting to \$8,164,762. This makes the total investment of the churches in classical institutions and seminaries to reach the large sum of \$22,837,162. Immediately connected with these halls of learning are some 700 of the church's finest scholars and most devoted Christians acting as teachers, while 14,343 of the best and brightest young men and women sit at their feet as learners.

Methodism has been a great educational force in this country. It took its rise in a university, and its leaders were trained in the oldest of English universities. The Methodist zeal for higher education has put her in the front ranks of the moral and educational forces of the age. Though among the youngest of Christian bodies of this country, the magnitude and extent of her educational work is second to none.

The Methodist Episcopal Church comprises less than one-half of the Methodists in the United States, yet she has 49 institutions of collegiate grade, with property and endowment of over \$17,000,000, and from the 6,000 students there are sent out annually 1,500 graduates with the Bachelor's degree. In 1892, she had 195 institutions of learning of every grade, with property and endowment valued at \$26,000,000, with 2,343 professors and teachers and 40,026 students.

"The increase in population in the United States from 1880 to 1890 was 26.7 per cent.; for the same period the increase of students in college classes in all schools in the United States was 53.1 per cent.; in all Methodist schools in the United States, 52.3 per cent." It is certainly a hopeful indication of the ambition and lofty purpose of Methodist youth that one-eighth of the whole number of students of the Johns Hopkins University are Methodists, seeking the broadest educational facilities. A church with such a record will not lose her hold upon the intellect and scholarship of the age.

Methodism has wisely undertaken to establish the American University in Washington City. The founding of such a university was the dream of Washington and other great statesmen. This is the most strategic educational center in America. The scientific and literary treasures of the government, aggregating a cost of more than \$33,000,000, and maintained at an annual expense of three and one-half millions of dollars, will be at the service of this university. The funds of the university will not be tied up in expensive buildings and equipment, but, like the great German universities, employed in paying enthusiastic professors of the broadest scholarship and culture to instruct graduate students in every department of learning, and to widen the horizon of knowledge. This is certainly one of the most magnificent opportunities in the history of the Christian Church to establish a powerful and comprehensive agency to help uphold and expand and organize a

Christian civilization. It will gain an increasing power through coming generations.

The Federal Government has, likewise, favored and materially encouraged the cause of education. The wisest statesmen believe that the colleges are not solely the auxiliary of the churches, but that they have an equal value to the State. They firmly believe that education is essential to the general good of the community, and worthy of favorable legislation. "During the first century of its existence, the United States made land grants for educational purposes of nearly 80,000,000 acres, a territory greater than all the landed area of Great Britain and Ireland, and more than half of all France. What a tribute to learning this munificence presents. Of these gifts it is estimated that more than 80 per cent. went to permanent funds for the elementary schools."

The spirit of the American people was shown in the Magna Charta of the Northwest, framed in 1787, which declared that "Religion, morality and knowledge being necessary to good government and the happiness of mankind, schools and the means of education shall forever be encouraged." In obedience to this spirit, the Federal government made grants of land to encourage and support institutions of learning, as follows: "One section of land in every township for common schools, and not less than two townships in every State for founding a university." Appropriations have since been made by the general government to establish and foster State universities. In 1862, the Morrill act was passed by Congress, whereby a liberal grant was made to provide for "the endowment, support and maintenance of at least one college, where the leading object should be, without excluding other scientific and classical studies, and including military tactics, to teach such branches of learning as are related to agriculture and mechanic arts, in such manner as the legislature of the States may prescribe, in order to promote the liberal and practical education of the industrial classes in the several pursuits and professions of life." This act was supplemented in 1890 by an additional provision of \$25,000 a year for the better equipment and endowment of each of the colleges of agriculture and mechanic arts. The land grant made by the general government to all the States aggregated 9,597,840 acres, from which was realized \$15,866,371.

The Hatch act of 1887 made generous Federal provision for the establishment of agricultural experiment stations "for the investigation of the laws and principles that govern the successful and profitable tillage of the soil."

The State universities numbered 30 in 1890, having 12,846 students and 964 instructors. The value of the grounds and buildings aggregated \$15,146,588, and the productive fund \$10,411,964. The total income for the State schools reached the handsome sum of \$2,176,250. These State universities have become fixed factors in our civilization, and give promise of accomplishing a great work for the people. What the character of the work shall be, remains with the American people to decide.

This century has witnessed in the United States the beginning and growth of *Colleges for Women*. This is the fruit of the increasing development of the idea and sentiment in favor of women sharing with men in the privileges of the highest culture and all rational enjoyment. Exclusive privileges and distinctions on account of sex are contrary to the character and genius of a free people. "If," says President Dwight, "education is for the growth of the human mind—the personal human mind—and if the glory of it is in upbuilding and outbuilding of the mind, the womanly mind is just as important, just as beautiful, just as much a divine creation with wide-reaching possibilities as the manly mind. When we have in our vision serious thought as the working force and end of education, the woman makes the same claim with the man, and her claim rests, at its deepest foundation, upon the same grand idea." The history of the movement in favor of the collegiate education of women is interesting and instructive. One of the first steps in this direction was taken by Mrs. Emma Willard, who opened a school for girls in Middlebury, Vermont, in 1808, which in 1819 was removed to Waterford, New York. Two years later she founded the Troy Female Seminary. Education for women received a new impulse through Miss Catharine E. Beecher, who, in 1822, opened at Hartford, Conn., an academy for girls, and it met with excellent success. Further efforts were made to extend education to young women of more mature years and give them the advantages of an intellectual training equal with that of colleges for men. The Wesleyan Seminary for women was founded at Kent's Hill, Maine, in 1821, and Granville College for women in 1834.

Through the earnest effort of Miss Mary Lyon, Mount Holyoke Female Seminary was incorporated February 10, 1836. The Elmira Female College was founded in 1855. These colleges multiplied rapidly and now there are more than two hundred institutions of higher learning devoted exclusively to the education of women.

Colleges for women have been quite liberally endowed by high-minded and generous individuals, and the stability and permanency of these colleges have thus been secured. Vassar College was incorporated in 1861. Mr. Matthew Vassar, the founder, gave 200 acres of land near Poughkeepsie, on the Hudson, which with his other gifts aggregated \$788,000. The total productive endowment in 1892 was \$1,018,000, and the value of the grounds, buildings, etc., was \$792,080 additional.

Wellesley College was founded by H. F. Durant in 1875, at Wellesley, near Boston. He gave 400 acres of land and an endowment of more than one million dollars. Smith College was founded through the beneficence of Sophia Smith, who gave \$400,000. Bryn Mawr, near Philadelphia, was opened in 1885, through the generosity of J. W. Taylor, M. D., whose gifts amounted to \$1,000,000.

In 1890, there were 179 colleges devoted exclusively to the education of women, having grounds and buildings valued at \$11,559,379, with scientific apparatus valued at \$419,000 more, and the productive funds aggregated \$2,609,661. The total number of students in these colleges for the same year was 24,851, and taught by 2,299 teachers.

The co-education of the sexes in colleges is also constantly growing in favor among those colleges which have given it the most thorough trial. Two hundred and seventy-two colleges in this country, or 65.5 per cent., excluding those devoted exclusively to the education of women, are open equally to both sexes. The favorable results as to scholarship, manners and morals of the two sexes have abundantly confirmed the wisdom of this method. The question of co-education has its complications, but with proper restrictions these are not serious. There is no more danger of women developing bold or masculine qualities of character in a college where co-education exists than in the high schools, or in social and business life outside of college. The charm and beauty of a lady are found in the qualities

of modesty and grace. The private life of the ladies attending a college where co-education exists is in most cases so regulated as to secure such home care and retirement as will help to preserve the charming qualities of womanhood. The ladies in these schools gain a certain poise and independence without boldness, which is of inestimable advantage. Aside from this they get a knowledge of character and life that is not likely to be secured in any other way.

The growth of the colleges since the war in the sixteen Southern States for both white and black population is very encouraging. Fully one-third of the colleges and universities and one-third of the instructors and students of the nation are located in the Southern States. Many of these colleges are only first-class academies, but they are doing an excellent service. Benefactions in behalf of higher education in the South have been something phenomenal in the history of philanthropic work. The Peabody Fund for education in the South was \$3,100,000. The Slater Fund \$1,000,000. Tulane and Vanderbilt each gave \$1,500,000 towards founding universities in the South. It is estimated that more than \$20,000,000 have been given by special donors for this purpose since the war. This vast sum has been augmented by the annual gifts of the churches for this object. The Methodist Episcopal Church had expended up to 1892 the sum of \$6,187,630.46 to promote higher institutions of learning among both white and black population in the South.

Other denominations have given largely in the same direction. These benefactions have given new impulses to the cause of education, which have been of vital importance in the regeneration of the social conditions of this section of the country. The annual outlay for schools in the Southern States increased from \$11,400,000 in 1878 to \$20,000,000 in 1888. All these educational influences have contributed to establish a New South that presages far-reaching possibilities for good for all time to come.

The growth, number and progress of the American colleges and universities is more and more attracting the attention of the civilized world. In 1890, they numbered 415, with grounds and buildings valued at \$65,000,000, with scientific apparatus and libraries valued at \$9,000,000, and the productive endowment funds aggregated \$75,000,000. The total income of these higher institutions of learning from all sources was \$11,000,000.

The colleges and universities and professional schools in the United States for the same year contained 135,242 students and 7,819 instructors. In the colleges and universities alone there were 46,131 men and 11,992 women. There were 34,964 in the normal schools, 6,349 in agricultural and mechanical colleges, and 35,806 in the various professional schools. Besides, there were 117 medical schools with 4,552 students, and 145 theological schools with 7,013 students, and 54 law schools having 5,518 students.

These facts give us some faint conception of the extensive educational agencies which have been provided, chiefly by private enterprise and by the churches, for higher education.

It is claimed by some that the number of colleges in this country exceeds at present the demand. It should be remembered, however, that we are building for a population that is likely to reach 500,000,000 people. There is no doubt but that the planting and expansion of colleges on a meager basis has been somewhat over done. The duty of the hour is for the American people to cease establishing more colleges, and to give their attention to strengthening those already founded, in order that they may increase their power and efficiency. The founders have planted better than they knew. The unfavorable conditions and sacrifice surrounding many of their beginnings strengthen the desire that these colleges may grow and flourish with each succeeding generation, and continue in their beneficent work of moulding Christian character and promoting human brotherhood.

III.

CHARACTERISTICS OF THE AMERICAN COLLEGE.

The American college occupies a distinctive place among the educational systems of the world. It differs from the English and Scotch systems, and is diverse in form and purpose from the German university system. The American college signifies more than the English *Grammar* school, the

French *Lycée* or the German *Gymnasium*, and its course of study is broader and more comprehensive. The German *gymnasia* hold the place of our high schools and academies, and their course of study carries the student through what is an equivalent to our Sophomore year in college.

The colleges established in the early history of our country were shaped in some measure after the English model, but the American college of to-day "is the bright consummate flower of democracy." We may apply to it what Lowell says of Lincoln:

"For him her old-world moulds aside she threw,
And choosing sweet clay from the breast
Of the unexhausted West,
With stuff untainted shaped a hero new."

The American colleges have held fast to the best of the ancient learning and utilized the best experiences and ideas of the English, German and French systems of education, and mapped out a distinctive system for themselves. They have sought to meet the needs of our age and the requirements of our generation, and we have as a product the modern American college, adapted to the wants of the people and the formation of a strong national character.

The American people believe in individual rights and personal sovereignty. They have accordingly shaped their institutions in harmony with this view. In Germany the man is educated largely for the State, but here we educate the man as a citizen and as an individual whose intrinsic dignity and value are worthy of training. The American college makes adequate provision for the full development of all the human powers and the exercise of the functions of the noblest manhood and womanhood. Her halls have always been wide open to all the youth of the land, who have gathered by the thousand to drink in "the American spirit of freedom and brotherhood of mankind, of reverence for God, for law, for the Bible and for the Sabbath." Our colleges have been built up through the generous and effective support of the several churches, and of the patriotic people. For more than two and a half centuries it has been the settled policy of the American people to maintain and perpetuate colleges. They are deeply rooted in the hearts of the people, since they are the offspring of their free-offerings and voluntary sacrifices.

A few unthinking people are indifferent and fail to see and realize the vital relations the colleges sustain to the national welfare; but the more enlightened public opinion is eager and restless for their advancement and influence. Our colleges are the pride and the crowning glory of the American people. They bring the nation more renown than all her fertile plains, rich treasures and splendid palaces.

In order to particularize some of the distinctive features of the American college, we need to understand our educational system as a whole. We start with the public school and impart to the youth a primary education. In the high school or academy the pupil is introduced into a higher circle of thought and life and then passes on to the college, where the aim is to extend general culture and prepare for special work. The educational system culminates in the university, which is devoted chiefly to technical and professional education.

These educational agencies do not differ in kind, but in degree. There is not as yet, however, a sufficient co-ordination of them to secure the greatest economy of time and strength in mental effort. The richest and broadest culture and scholarship demand a friendly and harmonious relation between all of these educational agencies. We are approaching co-operation and unity on these lines, but there are practical difficulties which it is hoped that time will help to solve. One of the difficulties has been that the standard of admission into many of our colleges has outgrown the capacity of the high schools. In order to supply the need of a more thorough preparation, a preparatory department has been maintained in many colleges. The present aim and tendency of our educational system is to introduce the pupil from the high school to the rank of Freshman in college. This condition can not become general unless there be a greater differentiation in the courses of study in our high schools. It is encouraging to see that in many States the high schools, academies and colleges are coming to a helpful understanding of each other's province, and that there is a practical agreement among them regarding a uniform minimum requirement for entrance into the Freshman class in college.

The prescribed *courses of study* in the average American college are broad and comprehensive. They cover the general field of knowledge. The regular

parallel courses of study are usually designated Classical, Scientific, Literary and Philosophical. These special arrangements aim to encourage thought and study along different lines. The groupings vary according to the time devoted to the study of languages and other special branches. Each of the courses includes the study of language, mathematics, science, mental and moral philosophy, and covers a period of four years, generally designated Freshman, Sophomore, Junior and Senior years. As a rule, in the Classical course the study of Greek and Latin is required, while Greek is omitted in the Scientific course, and more attention is given to the study of the sciences. The Literary and Philosophical courses substitute one or more of the modern languages for the ancient classics. The number of these courses may be multiplied indefinitely, especially in the universities where the grouping of studies is essential to the highest success.

The work of *the college and the university* so overlap each other that it is difficult to make clear their distinction. The word university is an elastic term in the United States, because until within a brief period we have had nothing more than colleges. Many of our colleges are called universities because of their chartered privileges, but their aim is to become universities in fact.

Hence the terms are often used interchangeably. The few universities we have are modelled largely after those in Germany and have grown up by a natural development out of colleges. The reverse is true in England, where the college has grown up within the university. The college originally signified a society of scholars. In this country it is an incorporated school of instruction in the liberal arts, having one faculty, with advanced courses of study.

The college and university differ first in their *aim*. The college endeavors to discipline the mind and form character for the broader work in a chosen field of university study. The thorough scholastic training is now regarded quite an essential preparation for the more advanced work of the university. On the other hand, the university aims at universal culture, and includes, if possible, every description of knowledge for the training of specialists in the various professions. Its aim is rather to do graduate work exclusively.

Again they differ in their *courses of study*. In the college, the courses of study include the higher branches of learning; and are so arranged as to give the student an outline survey of the field of knowledge. The study is largely restricted to preparing the student for his advanced professional and technical work. The university goes further and arranges its courses of study so as to supplement the instruction given in college and direct the student in an advanced grade of work in any department of intellectual life. The courses have the broadest scope and embrace departments in liberal arts, law, medicine, theology and science, each having a faculty composed of able professors. Gladstone gives the true historic idea of a university in these words: "To methodize, perpetuate and apply all knowledge which exists and to adopt and take up into itself every new branch as it comes successively into existence."

The college and the university likewise differ in their *methods of work*. The college seeks the highest results in discipline. Its method is more formal and didactic. In the later years of the college course a certain amount of specialization is usually allowed, both for the ends of discipline and as a provision for the work of the university proper. The university adopts methods of work along the line of original discovery, literary productivity, and the advancement of the kingdom of knowledge. The inspiring aim of the university is the discovery of truth. The student imbued with the spirit of research passes from the known to the unknown, and feels that he lives in an atmosphere of investigation, and in the center of the latest thought.

Finally, they differ in their resources. The college is usually limited in its means and appliances. On the contrary, the university, with abundant resources, great libraries and laboratories, affords a broader scope and wider opportunities for work and growth.

The *State and denominational colleges* have a common intellectual aim. The first of the two often have larger resources and aim to give more instruction in "practical affairs." Both State and denominational colleges are generous and liberal in their spirit and teaching. It is somewhat unfortunate that there should have arisen any occasion for criticism by the friends of either the State universities or of those under denominational control. One class of critics are ready to declare that the colleges and universities under

Protestant denominational control are sectarian. Whereas it is unfair to designate such colleges as sectarian, since as a class they are not founded solely in the interest of any single Christian sect and are not intolerant and bigoted. They set up no denominational standard for entrance, and teach no particular creed or dogma, but extend their privileges equally to all and on the same basis as the State universities. Hence, they are denominational, but not sectarian.

It is equally unfair to assert that our State universities are godless and run by political parties. The managers of them have possibly laid themselves open to this criticism because they often fail to recognize either the scientific bases or practical value of religion and do not permit it to rank equally with the other sciences in the courses of study. The right policy would not necessarily involve the teaching of religious dogma, but only of facts concerning man's spiritual nature, and the relative importance of the Christian religion among the religious systems of the world to meet the demands of man as a religious being. No reasonable man in a Christian nation should object to this recognition of the science of religion. The State universities should be at least religious in character without having any denominational bias. The teaching of dogma in our colleges for the sake of dogma would be narrow bigotry and rightly deserving of censure. The State universities are as likely to be open to this charge as the denominational colleges. The dogmas of scientists, politicians, legalists and physicians are as intolerant and engender as much strife as those of theologians. We are glad to believe however, that the dogmatic spirit in all lines of study is fast disappearing from our American colleges, and from the professions.

Again, the majority of the professors in the State universities are avowedly Christian. Possibly one-third of the State universities have Christian clergymen for presidents. After careful inquiry from those in a position to know, it was ascertained that in one of the oldest State universities there were eight professors out of more than one hundred who were unbelievers or skeptics, and in one of the youngest there were but three known skeptics among more than eighty professors. Even this small number should not be possible, because one "anti-Christian sophist or a velvet-footed infidel" may work moral and religious disaster to the young in any college. "A college," remarks President Gates, "must be either avowedly and openly Christian, or

by the very absence of avowed Christian influence it will be strongly and decidedly un-Christian in its effects upon students."

The State universities will gain greater influence if they will rigidly exclude from their teaching force the brilliant skeptic who "becomes the center of a coterie without his gifts, dazzled by his boldness, infected by his skepticism;" but rather employ Christian professors who will inspire a "noble ambition that unites in its scope the life that now is and that which is to come, that comprehends earth-born sciences and the philosophy of salvation, the tongues of men and the language of the city of the great King."

Likewise the State and denominational colleges and universities have the largest freedom and independence. Their boards of management are comparatively free from interference on the part of party politicians and demagogues, or of those influenced by denominational prejudices. Party leaders in the church or state may be equally liable to an undue bias or a partisan spirit and influence which is beneath the dignity of those who claim to represent the people in a Christian Republic.

The American college is a chartered institution, under the control of a *Board of Trustees* or *Regents*. These boards are composed of about twenty or thirty representative men in church or state. They are, in some cases, a self-perpetuating corporation, while others are chosen for a term of years by the affiliating conferences or synods. Occasionally, the Alumni of the college may elect some of the Trustees. The State universities are under a Board of Regents appointed by the Governor, with the consent of the legislative body, or are chosen by popular election. These boards meet once or twice a year. Their principal duties are to make laws for the government of the college; appoint the officers and professors, and fix their salaries and tenure of office, and hold all property entrusted to the college, and retain general supervision and control of all expenditures. These boards are the ultimate source of authority in all matters pertaining to the welfare of the college.

The Chicago University and some others have a *University Council*, composed of the chief administrative officials of the university. They direct all administrative matters. The *University Senate* is composed of the heads

of the departments of instruction. It is their duty to control all educational affairs. The *Harvard Corporation* consists of the President, five Fellows, and the Treasurer, with the right to fill their own vacancies. Their acts are "alterable" by the *Board of Overseers*, to whom they are responsible. This board consists of thirty-two members, elected by the Alumni.

The Faculty is a body of instructors. The universities may have as many faculties as there are departments of instruction. In the American college proper there is but one faculty, composed of all the instructors. It varies in number and efficiency according to the number of students and financial resources of the college. The proportionate number of professors to the students follows the custom of the best English and German universities, which usually is one professor for every twenty or thirty students. *The Dean* is an administrative officer of a department in a university, and is concerned with the internal discipline and executive affairs.

The Presidents of the American colleges are usually clergymen. They are chosen with reference to their pre-eminent ability as scholars and administrators. The President has oversight of the plan of instruction, the maintenance of discipline, and is the representative head of the college before the public. Considerable importance is attached to the office of the President, since the success of the college in a great measure depends on his individual talent and character.

The American college *professors*, as a class, may be characterized as having a living scholarship and a genuine speculative spirit, combined with tact and firmness in teaching. They are enthusiastically devoted to their work. There is a growing disposition to break away from mechanical and plodding routine, and adopt an intellectual, energizing style of questions in class work, that elicit enthusiasm and aid the student. Lecturing is but little used. The teaching is more of an active, earnest conversation on a special subject between the teacher and the pupil. The instructor seeks to lead, but not to carry, the student through the study. There is also less inclination to dogmatize, and the student's mind is trained to habits of original and philosophical investigation.

The students in our American colleges have been well estimated by Professor Von Holst in these words: "I have not only visited, but lived in a

number of countries, and the results of my observations of their higher educated youth is that, though by no means as to knowledge, yet as to the earnestness, steadiness and enthusiasm in the pursuit of knowledge, the American students stand first. And nature has not been in a stingy mood when weighing out their allotment of brains! Give them but the opportunities, and you will soon see whether they need to shun comparison with the scholars of any other nation."

College government is an important question. The college, as a distinct and separate community, has rules and regulations based on well-established principles, which aim to conserve the general good of the whole body of students. The college honor can not be sustained unless there is a recognition of authority and responsibility.

The college legislation and government rests principally with the faculty, overseers and trustees, who aim to be liberal, yet firm. College sentiment among students is often capricious and subject to sudden revolutions. Some of them have strong passions, immature judgments, and impetuous and weak wills, and authority must be lodged with those who will sacredly uphold law and exercise a firm, rigorous discipline.

In the early stages of college life in this country the regulations were quite severe. In many cases the college authorities did not hesitate to inflict upon the students corporal punishment for certain offenses. College Presidents would sometimes personally attend to the flogging of students, resorting to this punishment with great solemnity. Mr. George C. Bush tells us what occurred at Harvard College in 1674: "On that occasion the overseers of the college, the President and Fellows, the students who chose to attend having been called together in the library, the sentence was read in their presence and the offender required to kneel. The President then offered prayer, after which 'the prison keeper at Cambridge,' at a given signal from him 'attended to the performance of his part of the work.' The President then closed the solemn exercise with prayer."

Possibly this relic of severe college government found its example across the water, where it is related that in a bygone age a Fellow at Oxford, "who had been proved guilty of an over-susceptibility to the charms of beauty, was condemned, as a penance, to preach eight sermons in the Church of

Saint Peter-in-the-East." In the days of President Dunster, of Harvard, "no possible conduct escaped his eye. Class deportment, plan of studies, personal habits, daily life, private devotions, social intercourse, and civil privileges, were all directed."

The student should feel that, in disobeying the rightful authority of the college, he abridges the rights and privileges of every student. The college sentiment should be so strong against unworthy conduct that a student would as soon shrink from doing a mean action, and having it known, as any citizen outside the college community. When it is discovered that a student has mean and unworthy motives and wilful evil tendencies, he should be summarily dismissed.

In some colleges the students participate in the governing affairs. This is done by having representatives chosen from each college class, elected by their fellow-students, who unitedly compose a College Senate, with power to interpret the college laws, and deal with all questions relating to the good order and decorum of students. The President of the college is chairman, and has the power to veto the decision of the senate. There are many favorable features of this system. In the first place, it lessens the antagonism sometimes manifest between the faculty and students. There are no less requirements upon all college classes and duties, and it helps to remove any feeling of suspicion and the semblance of espionage. The students feel that they have been taken into confidence with the college authorities and will get strict, even-handed justice in college discipline. The result is that there comes to exist a more pleasant and friendly relation between the professors and students.

Again, this system gives the freest scope for teaching. The professor's time is not occupied doing police duty or sitting as a juror, but is given wholly to his work as teacher.

The self-responsibility of the student also has an educating influence, giving to the worthy and right-minded a better training for future citizenship. It is undoubtedly true that the autonomy of a college is an important factor in shaping the future liberties of our country. No college, however, can hope to uphold the highest standard of conduct by trusting to the force of rules and penalties. The spring of right action is in the heart. All college authorities

must rely principally upon appeals to calm reason and an enlightened conscience, reinforced by religious faith and feeling.

The general good order and morals of the students in American colleges are changing for the better. In a large proportion of our colleges only a small per cent. of the students use intoxicating drinks or tobacco. All reprehensible conduct must be carried on so secretly as to elude the college authorities. Those disposed to do evil represent only a very small proportion of the great body of students, but these give occasion for some supercilious and conceited correspondent of the public press severely to criticise the college government, and to give gross caricatures and exaggerated statements of the mischief done by this small percentage of students, and then include the entire academic body in the same general censure. It is generally believed by those qualified to know that the average morals and good conduct of the students in college are much better than those of the same number of young men outside the college community.

The chartered colleges are entitled to confer *degrees* as a measure of honor the college wishes to bestow on men and women of merit. This privilege has been so much abused by some colleges that a little confusion arises as to the true value and significance of the degrees conferred. In 1890, there were 8,290 degrees conferred in course or on examination, and 727 honorary degrees, by 415 colleges and professional schools.

In the best American colleges, the student completing the classical course receives the degree of *Bachelor of Arts* (A. B.)—*bas chevalier*, a knight of low degree; it signifies "inception in arts." If the student, after taking his bachelor's degree, pursues for a few years some literary or scientific study, he may receive the degree of Master of Arts (A. M.), meaning fitness to teach, a title which began to be conferred in the twelfth century. These degrees are granted as a reward of merit, based on examination and general fitness. The degrees of Doctor of Divinity (D. D.) and Doctor of Laws (LL. D.) are granted as honorary degrees to men of pre-eminent ability or for conspicuous services. The student who completes a college course or its equivalent, and follows it with a professional course in a university, receives a degree recognizing the fact. Schools of Theology confer the

degree of Bachelor of Divinity (D. B.) Schools of Law, Bachelor of Law (LL. B.), and Schools of Medicine, Doctor of Medicine (M. D.)

A post-graduate course of study, looking to the degree of Doctor of Philosophy (Ph. D.), has reference not so much to the professional and practical side of life as to the original investigation and exploration of a special subject, with no other immediate aim than the discovery of truth and a philosophical insight into the same. The student, before receiving the degree in the best universities, is required, at the close of his post-graduate work, to write a thesis which would be regarded as an original contribution to the subject discussed.

There is no practical uniformity in the scope and requirement of the work for this degree. The Doctor's degree should stand in this country, as it does in Europe, for research, and a general knowledge of philosophy, with ability to open up original sources of information. The student should be a resident graduate for at least one year, and after rigorous examination be required to contribute something to the advancement of knowledge, and withal be a man of good character and judgment, before receiving this most desirable degree in American and European universities. With such a uniform standard, this degree will not likely depreciate in public esteem, but have, as all degrees should, a uniform value. A federation of colleges may help to attain this end.

College degrees are not essential to a man's success in life, but when they are obtained as a reward of merit have a certain social value which usually insures a speedier entrance into any chosen field of work.

Another characteristic of American colleges is that they are *endowed* either by churches, by the state or by individual donors. The endowment is generally in the form of property or stocks yielding an annual revenue. It may be a sum of money given to the college, to be loaned and the interest to be permanently appropriated to the support of professors or applied to the current expenses. The amount necessary to endow a professorship varies from twenty-five to fifty thousand dollars. The fund thus given remains intact, and the interest or revenue of it alone is used to carry out the purpose of the donor.

No college of a high grade can exist without a generous endowment or aid from some source. Education in the colleges and universities throughout the world is given almost as a gratuity. It is maintained principally through the benefactions of wealthy men who erect buildings, found professorships and establish libraries for the use of others.

The resources of American colleges surpass those of any other country in the world. In 1890, the value of grounds, buildings and apparatus for 378 colleges in the United States was \$77,894,729, and the productive fund of 315 colleges aggregated \$74,090,415. In Germany, the twenty-two universities are national property, and are supported out of the national treasury at a large annual expense. The annual incomes of Oxford and Cambridge in England aggregate more than \$3,500,000.

Many of the American colleges have wealthy foundations. Harvard College has in grounds, buildings and productive endowment the sum of \$12,000,000, with an income in 1892 of \$978,881.92. Columbia College claims \$13,000,000, with an annual income of \$629,000. The estimated value of the funds of Cornell College is \$9,000,000, with an annual income of more than \$400,000, and Johns Hopkins University has \$5,000,000 endowment. In 1892, Yale College had \$4,019,000, with an annual income of \$520,246. The Northwestern University has nearly \$3,000,000 endowment and an annual income of \$225,000. Boston University has more than \$1,500,000 endowment and an annual income of \$160,000. Chicago University is one of our youngest universities, and yet it has in property and endowment \$7,500,000. These are only a small portion of the 415 colleges and universities in this country whose aggregate wealth and income are a source of satisfaction to all the friends of higher education.

The munificence of the wealthy men of this nation in behalf of higher education has excited the surprise and admiration of the old world. Within the last quarter of a century nearly seventy-five million dollars has been given for this cause. We recall with satisfaction some of these distinguished donors: George Peabody left \$6,000,000 of his estate to the cause of education; Isaac Rich, \$1,000,000 to Boston University; Johns Hopkins, \$3,140,000 to found a university in Baltimore which bears his name; Asa Packard gave \$3,000,000 to Lehigh University; D. B. Fayerweather left a

bequest of nearly \$3,000,000 to various colleges; Cornelius Vanderbilt gave \$1,000,000 to the Vanderbilt University; John C. Green gave \$1,500,000 to Princeton College; Amasa Stone, \$600,000 to Adelbert College; George I. Seney, \$450,000 to Wesleyan University; Matthew Vassar, \$800,000 to Vassar College for women; John D. Rockefeller's gifts to the Chicago University aggregate \$4,500,000, and Leland Stanford's estate will yield from \$12,000,000 to \$15,000,000 for the university that bears his name on the Pacific Coast. These men and a host of others will be remembered through succeeding generations for their generous liberality. The wisdom of these noble benefactions commends itself to the enlightened judgment of all good citizens. We believe, with President Schurman, that "the heart behind American wealth is at the bottom generous and discerning, and so long as money can foster intelligence, that heart will not suffer our civilization to become a prey to ignorance, brutishness and stupid materialism. No one knows better than the millionaire that man lives not by bread alone." The colleges are not founded to make money but to benefit the public by training and fitting men for the highest service. The majority of the students in American colleges are of limited means. If it were possible to sustain a first-class college by means of the income from students, the tuition would be so high as to limit the great advantage of a higher education to a few children of rich men. The annual cost of each undergraduate to the University at Oxford is \$700, at Cambridge \$600, and at Harvard \$300. If the actual expenses of running a college of high grade were divided proportionately among the students, they would have to pay three or four times the amount they now do for tuition. It is important that these educational advantages and incentives come within the reach of the humblest youth of the Republic, in order that they may be productive of the noblest manhood and womanhood.

Time and experience confirm the claim that the wisest and most permanent use of money is to help endow a college. Large wealth imposes obligations to make the best and most permanent use of it. Every man of means ought to be a patron of learning, because it yields the most satisfactory returns. "What better gift can we offer the Republic," says Cicero, "than to teach and instruct the youth." Wendell Phillips says that "education is the only interest worthy deep, controlling anxiety of thoughtful men," and President Gilman makes an equally forcible statement when he says that "to be

concerned in the establishment of a university is one of the noblest and most important tasks ever imposed on a community or on a set of men."

Many of our denominational colleges are parsimoniously sustained. If their constituency, both rich and poor, would become imbued with the spirit of the Colonial fathers, and arouse themselves to give liberally, their power and influence would be multiplied a hundred fold. "Let it not be forgotten," says President Thwing, "that if the college and university have large need of the wealth of the community, this wealth has yet a larger need of the college and university. Without the aid of the higher education in the past, much of the wealth could not have been created; and without the higher education of the present, wealth would now become sordid; gold-dust is no less dust because it is golden. The rich man needs the college as his beneficiary to help him to be a noble man quite as much as the college needs his benefactions to help it make noble men. A college in poverty can make men; a rich man (or a poor man, indeed,) cannot hoard in meanness without degradation of manhood." The colleges are the agencies to help call out the constructive talent of the nation. They open the pathway of opportunity to every young man and woman who desires to do the most for himself and humanity. Each one may link himself through his means and prayers to these powerful agencies for good.

IV.

THE FUNCTIONS OF THE AMERICAN COLLEGE—A SYMMETRICAL DEVELOPMENT.

The function of the American college is to train and develop all the human powers and faculties and help the student to attain a complete individuality. The broadest educational theory estimates the worth of all the human powers and has the highest notion of personality, the development of which demands the impact of physical, intellectual, moral, and religious forces. A rounded human development provides for the fullest and freest exercise of all the powers of being. "Culture," says Matthew Arnold, "is a harmonious

expansion of all the powers which make the beauty and worth of human nature, and is not consistent with the over-development of any one power at the expense of the rest."

Man is a unit, but inasmuch as God has endowed him with various capacities, his highest glory should be to develop them. The only limit to the college student is his native abilities and aptitudes, modified by the parental training, various social influences, and the preliminary discipline in the public schools. The college that receives the students, with their different aims and predilections and acquirements, and leads them to appreciate the greater possibilities of their natures, and arouses and encourages them to strive for their fullest development, is worthy of confidence and support.

A symmetrically developed manhood or womanhood implies *the training of the mind to think accurately and systematically*. The tried and historic conception of education is expressed in the Latin word, *educare*: to lead out. It is to draw out of the living soul, by the aid of books, appliances, and instructors, all its latent capacities, to help in the formation of correct intellectual habits, and pre-eminently to form character, and thus to enrich and broaden the whole range of life. The purpose of a liberal education is not to cram the mind with facts and principles, but "to build up and build out the mind" by the natural process of growth, so that all knowledge from without will be assimilated by a living mental organism. The important work of the college is to develop intellectual power. It is to aid in giving such a directive power of mind as will enable the student, by a fixed determination, to recall facts, apply principles, and perform acts as if they were spontaneous. It is so to train the judgment and reasoning faculties of the student that in the end he will have acquired power to do earnest intellectual work.

The direct aim of the instruction in college is to give the student access to vital and formative knowledge by studying man and his works, and nature and her works. He is thus led to know himself and to know the world, and the laws which govern nature, and man as a part of nature. He comes to see things as they are and to understand the laws of things, and thus he thinks and acts on more perfect knowledge. If the student is to be trained to

independent thought and action, he must have a sounder basis of knowledge than the teachings of those whose ideas and opinions are shaped by current, ephemeral literature. The majority of men act on too imperfect knowledge, because they will not take the time and exercise the patience to study the facts and principles relating to any given subject, and to do their own thinking. G  the says: "To act is easy, to think is hard." The remedy is found in the college courses of study which involve the study of ourselves through psychology, logic, and mental, moral, political and social philosophy, and the study of nature through the sciences and the laws of the world about us.

Another method, aside from the nature and scope of the studies pursued, to attain the end, is through the strong personality of the college professor. Alexander the Great said: "Philip gave me life, Aristotle taught me how to live well," and Emerson's judgment was that "it is little matter what you learn; the question is, with whom you learn." It is within the power of the college professor to help enlighten the understanding, strengthen and guide the intuitions and reasoning faculties, and to awaken within the student a consciousness of his new powers and capacities, and incite him to mental activity. The highest scholastic training demands that the professor studiously avoid all those methods of instruction which tend to mechanical habits of thought, and which check the mind's spontaneity of growth and repress the individuality so essential to true scholarship.

Incidental to intellectual culture in college is the ability to find promptly the information we want. "Next to knowing a thing," says Dr. Johnson, "is to know where to find it." No student can become a walking encyclop  dia, but he should learn while in college how to avail himself advantageously of reference books, libraries and other sources of information.

A college education likewise implies the ability to express one's ideas in a clear, appropriate style. The student should be able to tell what he knows. This clearness of thought and precision of expression is best acquired in the class room, in the literary societies, and in the classes devoted especially to the study of expression.

The intellectual aim of a college should be not only to awaken and develop independent thinking power as an abiding impulse which will prompt to effective intellectual work, but withal the will, the imagination, and emotive

nature should be so trained that the student will have a mental taste and moral appreciation for the best and noblest thought. Mental discipline and the dull routine of study will become cold and insipid unless the student is inducted into those fields of science and literature where he will find the richest sources of refined and elevating pleasures, and through them be incited to noble action. It is on these lines of study that the student acquires that spirit of study which becomes spontaneous, attractive, and joyous. He loves culture for culture's sake, and does not abandon its acquisition on leaving college.

A symmetrically developed manhood or womanhood involves *physical culture*. The ascetic idea of college life no longer prevails. The body, as well as the mind, is trained. The value to a student of good health and an alert and vigorous body cannot be overestimated. Educators are coming to realize more fully than in the past that the physical and psychical factors of life are inseparable. The body and mind are mutually related and affected. Systematic exercise stimulates quickness of mental processes and promotes brain power.

The leading American colleges are conducted on better physiological and hygienic principles than in the past. The student, on entering college, is subject to a careful physical examination by a competent physician, and a course of systematic physical training is prescribed. Any organic defect or incipient disease is discovered, and, if possible, corrected. Physical training has become an integral part of a good college course. Exercise is largely compulsory, because studious and ambitious students are likely to sacrifice physical for intellectual training.

A well-equipped gymnasium is essential for the most thorough physical culture. Bath-rooms, with facilities for plunge and shower baths, are an important adjunct in promoting that healthy condition of the skin which follows from frequent bathing. An athletic field for outdoor sports is, likewise, a valuable accessory to develop a lithe and active body.

The master of the gymnasium is generally a vigorous and enthusiastic instructor, who is able to conduct skillfully daily gymnastic class work, and relieve monotony and evoke interest by introducing a variety of exercises for the different college classes. He is also the hygienic adviser in all

matters relating to study and recreation. The students are taught that regular exercise, sufficient sleep, personal cleanliness, and proper diet will correct most of the so-called pernicious effects of over-study.

Outdoor sports, under proper restrictions, promote health and foster mental qualities. Foot-ball and base-ball have gained an undue prominence in some colleges. It is questionable whether they are the most desirable forms of exercise for physical development, since only a very small portion of the students at any one time can engage in them.

The evil features of inter-collegiate games, especially as practiced, offset their advantages. The undue excitement and spirit of rivalry fostered is foreign to the true idea of an earnest student life. The college is no monastery to make the student a recluse, but it should be a place of solitude, a modern cloister, where the student may be kept in partial isolation and away from the turbulent stream of public life and distracting social influences. The student may keep in the midst of the current of actual modern thought and life without sacrificing the quiet seclusion which is an essential requirement for the best scholarship.

These inter-collegiate games have been attended with temptations perilous to character. Abundant testimony is not wanting to show that their tendency has been toward rowdyism, gambling, debauchery, and other disgraceful conduct. Some of the games scarcely rise above the brutality of the prize fight. They have no elevating tendency, and no apology can be made for their roughness and bad moral effects.

The fine natural instincts of the majority of American people are repelled at such physical prowess. It is not necessary to introduce the element of pugilism in order to give vent to the superabundance of youthful animal spirits.

The abuse of these outdoor sports should not make us blind to the fact that they have a legitimate use. It is wiser to control and direct them than to curb the exuberance of good feeling which they call forth, and which might find expression in less appropriate channels. It should be borne in mind that all physical training is a failure unless the aim is to maintain and develop

health, to make the student symmetrical, strong, graceful and better fitted for the duties of living.

A symmetrical development involves, likewise, *the cultivation of the moral and spiritual nature.*

The Christian religion affords the broadest educational basis, because it presents the most exalted notion of personality and its development. It takes account of the deepest facts of our nature, and teaches philosophical principles that are true for all created intelligences. Hence it is that Christianity is essential to the best educational system. It precedes and governs true education. A narrow and false conception of man leads to building only one side of his nature. The will, the conscience, the emotional and spiritual natures demand a share in the broadest culture. We cannot divide these essential elements against themselves. The religious sentiment is so interwoven with our being that it cannot be eliminated or dethroned. It takes no subordinate place, because it is supreme. There is no true theory of life without the spiritual element. All theories of education and principles of action that do not recognize the relations of the human soul to the supernatural are out of harmony with the laws governing human life.

These truths have been impressed on the noblest minds. "The greatest thought," said Daniel Webster, "that ever entered my mind, is the thought of my personal accountability to God." And Channing says that "man's relation to God is the great quickening truth, throwing all other truths into insignificance, and a truth which, however obscured and paralyzed by the many errors which ignorance and fraud have hitherto linked with it, has ever been a chief spring of human improvement. We look to it as the true life of the intellect. No man can be just to himself, can comprehend his own existence, can put forth all his powers with an heroic confidence, can deserve to be the guide and inspirer of other minds, till he has risen to communion with the Supreme Mind; till he feels his filial connection with the Universal Parent; till he regards himself as the recipient and minister of the Infinite Spirit; till he feels his consecration to the ends which religion unfolds; till he rises above human opinion, and is moved by a higher impulse than fame."

The Christian religion is in harmony with intellectual activity, because it favors application to study, and enjoins the duty of seeking truth, as well as awakens and intensifies the love of the good and beautiful. In fact, the human intellect owes its greatest triumphs to Christianity. From the beginning, the Christian religion has assimilated and employed human learning, and has become a great formative force in modern intellectual movements. It favors a broad catholic spirit, and is the counterpoise and remedy of a narrow range of intellectual activity. History teaches that it has been a strong incentive in the search after truth, and the chief factor in training the race to a higher civilized life. The changes in the progress in modern civilization are stimulated and guided by Christian knowledge. The whole trend of modern thought and instruction in the higher intellectual circles is to apply Christian principles to the problems of life. In every age it has stimulated and invigorated the human mind. It has introduced nobler and better ideas of life, given impetus to self-development, and has produced the highest types of manhood and of womanhood. The inspiration and encouragement in advancing general intelligence and founding the higher institutions of learning is principally due to the Christian religion.

"From the days of the Apologists onwards," says Prof. John De Witt, "learning has always advanced under the fostering care of our religion. In the schools of Antioch and of Alexandria, in Carthage and Hippo, in the old Rome on the Tiber, and in the new Rome on the Bosphorus, throughout the period of the ancient church, religion is the great inspiration of intellectual labor. How true this is of the Middle Age I need not stop to say. Religion in Anselm assimilates the philosophy of Plato. In the Anglican doctor it employs the dialectic and metaphysics of Aristotle. And the true father of the inductive philosophy, who anticipated the Organon and the very Idola of his great namesake, is Roger Bacon, the Franciscan brother. It was to this wonderful and unique power of Christianity to assimilate and employ all the triumphs of the human intellect, that the Western World is indebted for the universities by which, most of all, learning was increased and transmitted from generation to generation. Bologna and Naples, the school of Egbert at York, the schools of Charlemagne in the New Christian Empire, with Alcuin as minister of education; the later universities, with their tens of thousands of eager students—Paris, Cologne, and Oxford—sprang into being obedient, indeed, to a thirst for knowledge, but a thirst for knowledge

which, in turn, owed its existence and intensity to the unique fact that Christianity alone among religions can assimilate and employ all the truths of human philosophy, of science, and of literature."

The importance of promoting religious culture in our colleges cannot be overestimated. Dr. Thomas Arnold has spoken words that should be preserved in letters of gold. "Consider," he says, "what a religious education, in the true sense of the word, is: It is no other than a training our children to life eternal; no other than the making them know and love God, know and abhor evil; no other than the fashioning all the parts of our nature for the very ends which God designed for them; the teaching our understandings to know the highest truth; the teaching our affections to *love* the highest good!" One of the greatest teachers, Mark Hopkins, on the fiftieth anniversary of his connection with Williams College, said: "Christianity is the greatest civilizing, molding, uplifting power on this globe, and it is a sad defect in any institution of higher learning if it does not bring those under its care into the closest possible relation to it." The profound French philosopher, Victor Cousin, declares that "any system of school training which sharpens and strengthens the intellectual powers without supplying moral culture and religious principle is a curse rather than a blessing." And President M. E. Gates says: "In place of the fermenting despair of nihilism, the reckless immoralities of atheism, and the suicidal negations of agnosticism which have cursed liberally-educated Europe, if we are to have here in America an influence strong, binding and beneficent in our social system, as the result of collegiate education, it must be, it can be only by retaining in that system a clear faith in God, and by making prominent, as the highest aim of life, the service of God in serving the best interests of one's fellow-men."

The goal of all education is fulness of stature of men and women in Christ. Art and science are a vain show without this aim. A man may have a brain as keen as a Damascus scimitar, and yet he is wanting without piety. This moral and religious equipment is necessary for right conduct which, Matthew Arnold says, is three-fourths of life. Other things being equal, the student that is touched and saturated with the religious life will be under the strongest motives and attain the highest culture and efficiency in life. A pure heart and a clear brain are closely related. "Our education will never be

perfect unless, like the ancient temples, it is lighted from above." Martin Luther said: "To have prayed well is to have studied well," which accords with the idea of the best scholars in former days at Cambridge: *Bene orasse est bene studisse*.

The Christian spirit is eminently favorable to culture and to the promotion of literary productivity. It helps to make brilliant and earnest teachers, and lends zest to professional ambition. "Other things being equal," says Noah Porter, "that institution of learning which is earnestly religious is certain to make the largest and most valuable achievements in science and learning, as well as in literary tastes and capacities."

President Gates forcibly expresses the thought in these words: "Man is not, and was not meant to be, pure disembodied intellect. True philosophy, as well as common sense, teaches that the heart and the will have their rightful domain in every man's life. If the understanding becomes arrogant and spurns the aid of the other powers of the mind, not only does the man become an incomplete man, but his intellect itself inevitably loses poise and clearness. The man ceases to be a man, and becomes a calculating machine, and his intellect becomes subject to those sudden reversals of legitimate processes and results which the law of construction for calculating machines renders inevitable in them, but from which *life* saves the living man, the feeling, worshiping soul."

There is nothing more important to equip the complete scholar and gentleman than the Christian religion. Tennyson's poetic interpretation of this truth is thus beautifully expressed:

"Let knowledge grow from more to more,
But more of reverence in us dwell,
That mind and soul, according well,
May make one music, as before,
But vaster."

The *methods of promoting religious life in college* are widely varied. One of the most effective means is the positive Christian faith and the personal religious influence of the college professors. The student enters college at a vital and perilous period of life. The judgment is often immature and the life

principles unsettled. In this speculative period the student may be blindly endeavoring to adjust his faith to his reason. Especially at this time he needs professors of superior reason, strength of faith and spiritual discernment to unveil the divine mysteries and aid in dispelling doubt. Ex-President Seelye, of Amherst, once said: "We should no more think of appointing to a post of instruction here an irreligious man than we should an immoral man, or one ignorant of the topics he would have to teach." It is certainly no narrow bigotry that leads the Christian public to demand that the colleges select professors loyal to the truth and the Christian Church. United with their scientific culture and professional ability as teachers they should embody Christian earnestness and purity of life, and aim to send out students with a positive and rational faith.

The parent who realizes that the moral character of his children will be fixed, in a large measure, while in college, believes that it would be moral suicide to permit them to come under the influence of a professor whose religious indifference, or unfavorable remarks about Christianity, might infuse the poison of skepticism, doubt, or indifference, and perhaps unsettle their early religious convictions, and "send them forth confused and adrift on the endless sea of conflicting notions."

The courses of study in college should be arranged so as to favor the study of the essential facts and truths of the Christian religion, and through them promote practical piety. There is no valid reason why the Christian religion, which is the chief energy and force in all intellectual culture, should not be distinctly and permanently recognized in the college curriculum. The well-established and accepted facts of the Christian religion should be gathered and studied with as much painstaking care, freedom of spirit, and loyalty to truth as the scientist studies his facts and constructs his theories. This method implies that the teacher and pupil hold in abeyance all those probable theories, speculations, and conjectures which are not established, as irrelevant to the work in hand. When this scientific spirit is more effectively introduced into the study of the Christian religion in our colleges, it will prepare the way for the restatement of doctrine so as to commend it with increasing force to every intelligent student. Christian truth is capable of being built up into a system as scientific as any other. The professor, in leading the earnest student in search of spiritual truth, will

exercise tolerance and tact, so that he will not awaken suspicions of being actuated by a narrow bigotry, or appear as a lover of dogmatic teachings.

Again, it is better to select text-books that have been written by capable men who are in sympathy with the Christian religion. The student with an immature mind, who seeks to build his faith and theories of life on the teachings of those whose predilections are away from Christianity, will find it fatal to his lofty ideals and aspirations, while instruction based on Christian theism tends to lift the mind upward, and to foster a hopeful and earnest moral and intellectual life.

We grant that Christian character can only be incidentally produced through the subjects studied. The same study may be taught in different ways, and with entirely different results. The intellectual processes involved in study do not necessarily exert a spiritual influence. The aim and spirit of the professor and student will determine whether the study pursued shall contribute to the cultivation of greater reverence and exaltation of the soul. The charm of scientific study may so occupy the student's attention as to exclude all thoughts of the spiritual and eternal, or he may "look through nature up to nature's God." The student may be so absorbed with the human events and material conditions of history as to overlook the light of God's presence and guiding hand in it all.

To be liberally educated in Christian America, one should have a knowledge of the English Bible. It is the fountain and conservator of pure English and the storehouse of the most inspiring thought. Its classic beauty and lofty speculations and sublime morality are essential to a liberal education. "Froude calls the Bible the best of all literatures. Daniel Webster read the Bible through every year for its effect upon his mind. Charles Sumner kept the Bible at his elbow on his desk, and could find any passage without a concordance. Great men have found the Bible a great inspiration. But not this alone—as a great and inspiring literature,—but as a source of spiritual life and power, the Bible is the basis of true collegiate growth."

The study of the English Bible in colleges is important in developing the will and the conscience, and in evoking religious feelings which have a practical influence on conduct. It certainly imparts a vigorous character to education, and brings men face to face with the facts of sin and its remedy.

The presence of Christianity in the intellectual life of the student is corrective of selfishness and other vices which enslave the intellect and render life a disastrous failure.

It is encouraging to note that the study of the Bible is finding a place in the American college curriculum on a level with other studies, and time is allotted to attain a certain intellectual mastery of it. The active class instruction is as exacting and exhausting as any part of the college course. The student is led to trace the historic movements and to perceive the organic character, the literary forms and personal factors in its composition. The inductive method adopted develops original and independent students of the Word. The intellectual, devotional, and practical ends attained by this study are a powerful factor in upholding and maintaining the moral and spiritual character of the students.

Another method is that of *religious worship*. Students living in a community with a separate intellectual and social life should be required to meet daily for religious worship and instruction. The sacred moments spent in the college chapel by the whole college community are an appropriate recognition of the worth and power of the Christian religion, and do something to meet the spiritual needs and aspirations of the human soul. The daily gathering of the academic body to listen to a brief but suggestive exposition of scripture, and to unite in praise and prayer, cultivates reverence and devotion in the student, and will be regarded by many of them in after years as among the most delightful experiences in college life. If the religious services are not made perfunctory, but attractive and inspiring, in college, the students may pass to the university in their maturer years with devotional habits, and, likely, to avail themselves of its voluntary system of daily religious exercises.

The colleges should ever keep in view the original aim of the founders to make them centers of evangelical power. Piety, however, should not be a substitute for honest scholarly work. They should never permit their enthusiasm for an intellectual training and the growth of the sciences to obscure or conceal Him who is the Light and Life of all men. Their immediate and primary aim should be to promote intellectual culture, but this in nowise involves a departure from the spirit of the forefathers who

made them agencies for defending and propagating the gospel, and for leading the youth to remember that "the fear of the Lord is the beginning of wisdom."

It is evident, then, that the function of the college is to unfold the intellectual, physical, moral, and spiritual life of the young people, and especially to form character that shall be fully equipped for carrying out the divine purpose of life.

THE ADVANCEMENT OF KNOWLEDGE.

Another function of the American college is to extend the objective field of knowledge. The enlarged range of knowledge in our day is owing principally to the clear thinking and earnest, original, productive work done by college professors and students. They have done more to extend the empire of thought than any other class of intellectual workers. The college is the home of the arts and sciences, and it exists to teach and promote them. Professors should have the ability and the time, more and more, to make investigations, to extend the domain of truth, and to give philosophical and scientific guidance to the nation.

The university proper, as now being developed, regards as its special function the training of men for research and professional work. Its ample facilities and its methods of work give advanced students rare privileges in any department of research.

"The modern university," says Professor Josiah Royce, "has its highest business, to which all else is subordinate, the organization and advance of learning. Not that the individual minds are now neglected. They are wisely guarded as the servants of the one great cause. But the real mind which the university has to train is the mind of the nation—that concrete social mind whereof we all are ministers and instruments. The daily business of the university is, therefore, first of all, the creation and the advance of learning, as the means whereby the national mind can be trained."

The constructive intellectual spirit so paramount in the university begins in the college. The more formal methods of disciplinary work at the beginning

of a collegiate course gradually shade off, during the closing years, into the methods and spirit of original discovery adopted in university work. In the college there is kindled in the student the love of new truth and an enthusiasm for the advancement of learning. He is led to undertake creative work, and become an active, intellectual producer, with aspirations to widen the horizon of thought and weave the best results of his discoveries into the warp and woof of the social organism.

The steps leading up to the important period in the student's life where research is for the sake of fruitfulness are traceable in the historic development and requirements of college studies. In nearly all the colleges there is manifest a growing spirit of freedom in pursuing a course of study. There is little doubt that elective courses of study are a recognized necessity and benefit. It remains, however, an open question what studies should be required and what elected, and when the work of specialization should begin. If we keep in view the fact that the primary aim of a college education is to elevate and broaden the student by training him to clear and exact thought and accurate observation and expression, we will see that, whatever the course or subject of study chosen, it is only the means to this end.

Required studies should be based upon the principle of the instrumental, substantive and interpretative elements in a liberal education. For example, the study of language is important as the instrument of thought. A knowledge of the rich and copious foreign languages opens up the wisdom of the past and present, and their study develops memory and precision, as well as stimulates and provokes thought. A knowledge of some of them is essential to the highest professional success. The student who can read and appreciate the foreign languages and appropriate their contents has a decided advantage.

Mathematics is, likewise, an instrument of thought. It is the foundation of the physical sciences and the framework of the material universe. Its study trains the mind to think in relations and quantities, and helps to obviate loose and confused thinking. Logic and psychology are also important factors in developing the power of orderly and protracted thought.

The substantive element in a liberal education is found in the study of the natural and moral sciences. The study of them is both attractive and stimulating, and helps to store the mind with useful facts and principles. A general study of science should be required. A knowledge of any favorite science involves in some measure a knowledge of others. Physics, chemistry, biology, astronomy, are all more or less related. There is an interacting and interweaving of the facts and principles. Aside from the information imparted, there is no other class of study that will so effectively train the mind to accurate habits of observation.

Philosophy is the interpretative element in education, and helps to give unity to our knowledge. No one can reasonably lay claim to be liberally educated who has not some knowledge of the philosophical principles which underlie and explain the phenomena of history and life.

These required studies should be embraced and upheld in all college courses in order to give unity and consistency to the knowledge of the student. The value of these different studies cannot be reasonably doubted. The colleges of the past developed strength by studying these few subjects. No technical studies or professional training can be substituted for this scholastic training. The professional man especially needs this general culture, in order to escape the danger of concentrating and contracting his intellectual interest. Colleges may vigorously adhere to these scholarly requirements, and yet advantageously introduce the elective system. The student must have depth as well as breadth of scholarship. This can be effectively done by the specialization which the elective system affords. The character of the different studies chosen, however, should have a cohesive and logical connection in order to secure concentration and attain the best results.

The student who has had the advantages of a thorough preliminary training for admission to college, and has done faithful work in the required studies of the Freshman and Sophomore years, should have acquired such mental discipline and reached such a plane of scholarship that he is prepared for the more advanced work in special studies looking toward his life work. He should then be allowed to choose, within reasonable limits, those subjects for study during the Junior and Senior years in which his natural aptitudes

and modes of thought would lead him to seek the highest degree of proficiency. This plan accords with the German system of education at the point where the student leaves the required work of the gymnasium and enters upon the elective work of the university. The most aggressive colleges in America have adopted this method, and are satisfied with the results.

The elective system is beset with difficulties. Liberty is always subject to abuse, but the best attainments are found where negligence and mental trifling are possible. The advantages, however, are many. When the student decides upon a course of study suited to his real or imaginary needs, he exhibits more enthusiasm than if it is imposed. He is spurred on to his best effort, and develops personal power in original work. He gains depth and breadth of training, and is better fitted for more extended study in a university where the means and facilities are unlimited for the highest attainments in technical and professional training.

This is the sure way to raise up a class of experts and investigators who will keep in touch with the sources of knowledge, and, by doing original work, contribute something new that will widen the horizon of knowledge and extend the empire of thought.

PREPARATION FOR SERVICE.

The function of the college is something more than developing men and women and promoting knowledge. Its aim is, likewise, *to prepare the student for service*. The knowledge and culture gained in college are only a means to an end. The student must not only know something, but be able to do something in the sphere of life. The ultimate object of all culture is to equip a person for life's work. Milton declares that the proper system of training is "that which fits man to perform justly and skillfully and magnanimously all the offices, both private and public, of peace and war;" and Herbert Spencer says that "the function which education has to discharge is to prepare us for complete living." And again, "the great object of education," says Emerson, "should be commensurate with the objects of life." The mind, placed in actual conscious relations with existing realities and phenomena, should be prepared for the largest service. To know, see,

and learn the truth is a preparation for doing. The high type of manhood and womanhood which a liberal culture in college aims to promote should fit the student for every walk of life, in the family, society, church, and state.

The purpose of a college education should be twofold—*professional* and *humanitarian*—to prepare for one's vocation in life, and to cultivate humanitarian sympathies for the largest service. A person possessed of the humanitarian spirit realizes that the individual life is rooted in God, and consequently has a broader and deeper sense of human brotherhood, which enables him to keep in vital and sympathetic relation with human activity and experience. When these two aims blend, the best results are obtained, both for the individual and the community.

Aside from the scientific passion for knowledge, there is a view of culture, as Matthew Arnold puts it, "in which all the love of our neighbor, the impulses toward action, help, and beneficence, the desire for removing human error, clearing human confusion, and diminishing human misery, the noble aspiration to leave the world better and happier than we found it—motives eminently such as are called social—come in as a part of the grounds of culture, and the main and pre-eminent part."

It is to be feared that in some colleges the ideals and spirit are such as to lead the student to place power on wealth above culture, and social position above usefulness. Professor J. M. Hart estimates that nearly one-half of the students who attend Cambridge and Oxford Universities, in England, do so not for the sake of study, but in order to form good social connections. Liberal culture should not be sacrificed to preparing men for idle social life and paying places. Colleges do not exist to train the students' powers for personal benefits, but to promote culture, to the end that a larger service may be rendered to human progress. "An education," says President Hill, "that fails in producing lofty character, sustained and nourished by a pure faith, may, indeed, fill the world with capable and masterly men in their vocation; but, unless it can soften the heart of success and open the palm of power, it only strengthens the grasp of greed, and misses the making of noble men."

The true conception of man and his duties leaves but little room for individualism or insolent self-assertion. No one can divorce himself from

his fellow-men and their interests without lowering and debasing his own vocation in life, and becoming enfeebled and stunted in his own development. "The supreme object of the college," says President M. E. Gates, "is *to give an education for power in social life*." Every advancement in knowledge should tend to strengthen the bonds of human sympathy. Learning should be turned to the advantage of the people, and thus cause intelligence and helpfulness to go together. The great example of Christ teaches that a life of service is the only real human life. The quality of the student's character will be determined by his use or abuse of opportunity for service.

The very character of culture is social and beneficent. The great men of the world have most fully represented humanity. Touching the hearts of men, they have brought out the best of humanity in themselves, illustrating the truth of the divine law whereby we attain eminence, "Power to him who power exerts." The best thought not only contributes to the fulfillment of duty, but we receive impulse and mental activity by obedience to duty. Farrar says: "There are some who wish to know only to be known, which is base vanity; and some wish to know only that they may sell their knowledge, which is covetousness. There are some others who wish to know that they may be edified, and some that they may edify; that is heavenly prudence. In other words, the object of education is not for amusement, for fame, or for profit, but it is that one may learn to see and know God here, and to glorify Him in heaven hereafter. Our education is desired that, in the language of a Harrow prayer, we may become profitable members of the church and commonwealth, and hereafter partakers of the immortal glories of the resurrection." The measure and worth of a college should depend upon the pure and forceful character manifest in its students, and upon their willingness to employ the ability and knowledge acquired to serve the highest good of their fellow-men. The college that does this most efficiently will produce the best results.

When this conception of the function of a college is more thoroughly fixed upon the attention of educators and students, it may help to present in a clearer light some educational problems in regard to culture and practical training in college. On the one hand, there is a demand that the work of our colleges should become higher and more theoretical and scholarly, and, on

the other hand, the utilitarian opinion and ideal of the function of a college is that the work should be more progressive and practical. One class emphasizes the importance of true culture and of making ardent, methodical, and independent search after truth, irrespective of its application; the other believes that practice should go along with theory, and that the college should introduce the student into the practical methods of actual life.

They are both, in a measure, right. There are forces at work in society to strengthen the demand that colleges teach the branches of industry, as well as prepare men for the so-called learned professions. The demand is based on the worth and dignity of intelligent labor. In fact, a scientific and technical education in some branch of industry has already won its way to the rank of a learned profession.

The demand for industrial education has grown out of a reorganization of the industries and trades of the world. The great industries of the country require men of trained minds and directive intelligence to organize and control them. Mechanical skill is in great demand, and workmen must be trained not merely in dexterity and skill in the use of tools, but they must be so instructed in the principles governing science that they shall be able to reach results of the highest practical value in the sciences and arts. This age requires better mechanics, manufacturers, foremen, architects, farmers, and engineers—men whose creative genius will help to awaken the aspirations of the race to master the forces of nature and bring in an era of more convenience, comfort, and leisure for the cultivation of the mind and heart.

Our systems of education are planning to meet the needs of the people. Manual training that is adapted to youth between twelve and seventeen years of age is incorporated in the curricula of many of the existing public schools. Besides, we have in the United States more than one hundred advanced schools in technology founded as independent organizations. One-third of them have shops for laboratory practice.

The fact that such a prominent place has been given to the physical and practical sciences in the courses of study in colleges shows that these institutions are responding to the constantly increasing demands of a practical age. Scientific departments have been advantageously established

in connection with our well-endowed universities. It is both desirable and practicable to give instruction in mechanical, electrical, and civil engineering in our high grade colleges. This should not be done, however, at the expense of liberal culture.

How far the colleges can meet the demand for technical and practical education depends upon their condition and resources. They cannot make bricks without straw. Wealthy men cannot perform a more generous act than to help establish these schools of technology in connection with our colleges, in order to give instruction in the practical and useful arts of life.

There is danger, perhaps, in pressing the utilitarian principle in education too far. It is not the colleges that make the greatest show of utility that develop the most effective men. In the effort to secure a practical education, it is important not to lessen the power to understand and apply the foundation principles which underlie actual practice.

In the German universities the practical and technical are left alone. Professor J. M. Hart says of them that their "chief task, that to which all their energies are directed, is to develop great thinkers—men who will extend the boundaries of knowledge." We are under different conditions in this country, but the importance of the principle should not be overlooked. Every one has not the desire or ability to be a great scholar and thinker, but preparation for all the so-called practical careers of life should at least carry the student through the rigorous discipline of a college course up to the Junior year, when he may elect studies of a more technical nature looking to his life work. This is the best way to get a profound insight into principles from which to deduce practice and promote the interests of human society.

Professor Josiah Royce has well said that "the result of this 'conflict' between the two ideals of academic work has been the union of both in the effort of all concerned to build up a system of university training whose ideal is at once one of scholarly method and of scientific comprehension of fact. For the scholar, as such, be he biologist, or grammarian, or metaphysician, the exclusive opposition between 'words' and 'things' has no meaning. He works to understand truth, and the truth is at once word and thing, thought and object, insight and apprehension, law and content, form

and matter. * * * There is no science unexpressed; there is no genuine expression of truth that ought not to seek the form of science."

The importance of scientific theories leading to the best practical results is illustrated in the case of Columbus, whose investigations led him to believe in the sphericity of the earth and the probability of land in the far West. "Adams and Leverrier discovered Neptune simultaneously and independently, simply because certain observations had revealed perturbations that could be most naturally accounted for by the existence of an unknown planet." After Professor Helmholtz and others had made known the subtle laws of the transmission of sound, there was only a step to its practical application in the use of the telephone.

The essential condition in all industrial and social progress is the acquisition of judgement, skill, and foresight by patient study of facts and principles. It is energy within the being that gives birth to achievement in the outward sphere of practical life. It is certainly the prerogative of the colleges to extend the best educational opportunities to the people. It should embrace their intellectual and industrial pursuits.

The lofty and sacred purpose to render the highest service, to advance the welfare of men, is best reached by training men and women for leadership. The demand for educated and influential Christian leadership is greater than the supply. In 1890 there were about 15,000,000 pupils in the public schools receiving elementary instruction, while only one in 455 of the population was under superior instruction in colleges. The majority of this small number will be among the real leaders of the country. The character of the nation will, in a large measure, depend on the character of the colleges which train and shape these leaders.

A comparatively few men act as leaders, frame platforms, and shape legislation. It is quite difficult to find even this small number who are qualified for leadership. Nearly all our political and social reform movements are asking for a Moses, or a Luther, or a Lincoln, to lead them to victory. Some organizations of labor are officered by foreign born leaders who are ignorant and out of sympathy with the moral ideas and principles that have shaped our national life. There is a large number of imperfectly equipped men in all professions and in social movements, presuming to act

as leaders, who might well be replaced by disciplined and cultured men, able to grapple with modern social problems, and to conduct the people to higher thought and nobler action. Men who are to become leaders and gain a strong hold on society must have a good foundation of general knowledge, and be trained to think on complicated questions. The man of thorough training, whether literary, scientific, or practical, has an immense advantage in leadership.

It is the prerogative of the college, in its aim to serve the people, to extend such educational opportunities to youth as will equip them for true leadership in every vocation of life.

The American college student should be sent forth with a purpose even stronger than that of the Greek youth, who took the oath of citizenship in these words:

"I will transmit my fatherland [its institutions, its civilization, its system of education, its people], not only not less, but greater and better, than it was committed to me."

V.

STUDENT LIFE IN COLLEGE.

Admission to college is dependent upon the mental and moral fitness of the student. If the student has completed the work of an advanced high school, or that of an academy, he may in many colleges pass immediately into the Freshman year without examination. The student is generally required to have, as a necessary preparation to gain admission to the Freshman class, three years of Latin and two of Greek, or an amount of modern languages equivalent to the Greek, besides mathematics, history, and English. In some cases the qualifications of the candidate must be such as to enable him to read at sight either Greek, Latin, French, or German. An essay in English must be correct in composition, spelling, grammar, expression, and division into paragraphs.

Some favor admitting the student on trial, and giving him an opportunity to show his fitness and worth by application to study. Certainly the best test of the student's knowledge is the ability to pursue advantageously the prescribed course of study.

After admission to college the student has at least fifteen hours per week of class room work. He may select, within a limited range, his studies. This selection is done under the guidance of the professors, and depends largely on the acquirements or deficiencies of the student. About three-fourths of the Freshman and Sophomore years are devoted to the classics and mathematics. A large share of the work in the Junior and Senior years may be devoted to specialization in science, language, mathematics, history, sociology, or philosophy. In some cases elocution, music, and the fine arts rightly receive a fair share of attention on the part of a large number of students throughout the college course.

The advantages of a college education do not consist alone in the training of the faculties and the acquisition of knowledge, but one of its chief advantages grows out of the incidental noble and generous associations and influences.

The college is a homogeneous community of a distinct and peculiar type. It is a little world by itself. The professors and students are separated from the common activities of life, and a common feeling unites all in a common bond. There are poured into this community the hopes, aspirations, habits, and tastes of the different students, which are soon molded into a common life, and become, in turn, an important factor in forming the character and directing the life of the student.

The college classes become the organic centers of college life. For four years the students meet, at least in the smaller colleges, in the same lecture rooms for common studies, and become acquainted with each other's talents, tempers, and characteristics. It is within this charmed circle that the students find their associates and form warm and lasting friendships. It is not to be wondered at that class spirit runs high and class sentiment becomes a strong abiding power with the student. It is worth much to any young man or woman to be initiated into this hallowed sanctuary and catch its spirit and receive its uplifting influence. These central forces of the college classes naturally combine into a community with a common life. Thus each college comes to have a *genius loci* of its own. The subtle and fascinating influence of the common life and spirit is the *esprit de corps* of a college, and exerts no small influence over the life of the students. It gives exhilaration and stimulus to the students, and its formative power is felt throughout their lives, molding character and giving form to their opinions and direction to their aims, so that the college becomes a real *Alma Mater*. It is this spirit that makes and enforces a peculiar sentiment in the college community, which becomes almost as strong as positive law.

These influences emanate in various ways. No one can trace them to their ultimate source, but all feel the effect of these dominant forces, and realize that their lives are, in some measure, gradually but surely becoming molded and shaped by them. These influences are among the most cherished recollections in after years, and unite the student to his college with affectionate regard. There is certainly no better place for our youth to form and solidify a manly character, and develop independent convictions and humanitarian sympathies which will be of lasting satisfaction.

Noah Porter, in speaking of the benefits of association in a college community, truthfully says: "It is enough for us to be able to assert that thousands of the noblest men, who stand foremost in the ranks of social and professional life, would be forward to acknowledge that they are indebted to the cultivating influences of college friendships and college associations for the germs of their best principles, their noblest aspirations, and their most refined tastes. * * * True manhood, in intellect and character, is in no community so sagaciously discerned and so honestly honored as in this community. Pretension and shams are in none more speedily and cordially detected and exposed. Whether displayed in manners or intellectual efforts, conceit is rebuked and effectually repressed. Modest merit and refined tastes are appreciated, first by the select few, and then by the less discerning many. Each individual spectator of the goings-on of this active life is learning intellectual and moral lessons which he cannot forget if he would, and which he would not if he could, and he comes away with a rich freight of the most salutary experiences of culture in his tastes, his estimates of character, his judgments of life, as well as of positive achievements in literary skill and power."

Some of the effective means of social life among the students are the *open* and the *secret* societies. They are purely voluntary, and are originated and managed by the members.

The *Greek Letter Societies* are *secret*, and prevail in nearly all colleges. They are generally limited to ten or twenty members, and the chapters in the different colleges bear a friendly and mutual relation to each other. Among the Eastern colleges, nearly all these societies have elegant chapter houses, in which the members have rooms, and where they enjoy homelike comforts; while in the Western colleges the societies have attractive rooms, with tasteful appointments, which become a place of rendezvous for their members. Their only bond is congeniality. Some very different types of character are manifest in these societies. Students group themselves according to their common tastes, habits, and character. Some societies aim at scholarship or literary excellence, while others make wealth or social qualities an essential requirement. Even "fast fellows," if there be such, are eager to group themselves together into a secret society. A few of these societies are of a literary character, but the object of the majority is to promote sociability. It is claimed that their influence in some colleges is positively injurious, while in others they are beneficial and helpful in cultivating social qualities and in establishing warm intimate friendships among the members.

It is a question whether the attendant evils do not offset their advantages. They are expensive, and often accompanied with distractions unfavorable to student life. Sometimes the late hours and suppers and other convivial indulgences absorb time and lower scholarship. They afford opportunity secretly to do evil. The members may plan escapades and hatch intrigues, and cover them up so as to make it almost impossible for the college authorities to discover the guilty ones. Yet many excellent things are said of them and of the mutual benefits to their members.

The *open* societies, devoted exclusively to literary work, need no justification. They are voluntary associations for general literary and forensic culture. Oratorical and literary accomplishments are a prerequisite to the highest success and usefulness. The student who improves the opportunities of these societies need not neglect his regular college work, but in them can train himself to think consecutively, and gain facility of expression and an acquaintance with parliamentary law. If he makes faithful preparation, he will escape bombast and loose thinking and expression, and will become familiar with public movements, political questions, and social tendencies. For these and other reasons the literary societies should be encouraged, and students should consider it a privilege to become members of the same.

Political clubs are, likewise, organized among the colleges to promote the success of their several parties and the triumph of their respective principles. At the time of national contests the clubs are especially active at

mass meetings, in joint debates, and speeches, which set forth the merits of party principles and candidates. These experiences are both pleasant and instructive. The dignified participation of students in active political work tends to fire their patriotism and better equip them for the important social and civil duties of life. Political leagues are now organized in nearly all our colleges, with a view to strengthen the political party ties of the students in the several colleges and extend the party spirit and principle.

Glee clubs and other musical clubs, together with classical and scientific clubs, likewise afford ample opportunity for cultivating social life, and furnish pleasant entertainment.

Interest in athletic sports and outdoor amusements is often intense. Foot-ball and base-ball are the most popular games. Boat clubs are especially popular at Harvard and Yale. Bicycle clubs and lawn tennis clubs are made quite enjoyable to a large class of students.

College students also edit and publish college newspapers and journals. They are issued as daily, weekly, or monthly papers, and are supposed to voice the sentiment of the college and reflect its social, intellectual, and moral conditions. These journals help to keep the alumni and the undergraduate students in touch with the college and its work.

The religious life in college is very important. One of the primary purposes of the founders of American colleges was to promote such a religious life among students that they would go forth into all vocations as religious teachers and leaders of the people. This religious purpose has not been entirely thwarted. The general religious interest was never more marked and aggressive than at present. From one-half to five-sevenths of the students in American colleges make an open confession of Christ. In 1893, there were 70,419 young people in Protestant colleges. Of these, 38,327 were members of churches. Within the last few years the religious tone of our colleges has been elevated and improved. The average American student feels the need of educating the spiritual nature, and that there is no better way to attain this end than through a knowledge of the Bible and the soul touch of the Christ-life.

College authorities, recognizing the student's need of daily spiritual food, almost universally require once a day attendance at college prayers, which last from fifteen to thirty minutes. The students have frequent opportunities to meet the college pastor or one of the professors for conversation on personal religion.

Revivals are of frequent occurrence in many of our American colleges. These religious awakenings are strong and pervasive, and not only show the deep religious interest, but give a Christian tone to the body of students. The extent and intensity of these revivals in some colleges is so manifest that from three-fourths to nine-tenths of the graduates go out from their halls professing Christians.

The Young Men's and Young Women's Christian Associations are organized in nearly all the colleges, to secure growth in the Christian life and to encourage aggressive work among the students. They have either separate buildings on the college campus, or rooms fitted up in some of the college buildings, for their regular religious meetings. These associations are operated through standing committees, composed of one or more members from each college class. These societies have done much to awaken, increase, and intensify the interest of the students in religious matters, and by prayer and mutual sympathy have strengthened each other's Christian character and principles during the trying years of college life.

The morals of students should not be expected to rise much above the morals of the homes from which they come. The formative period of the student begins prior to college life. Parents who neglect this opportune time for training the moral life should not place this responsibility upon college professors and expect them to make up for parental neglect. It is a well-known fact, however, that only a very small per cent. of college students are known to be immoral. The prevalence of the drinking habit is decreasing. In one or two of the Eastern colleges a large per cent. of the students will take a social glass on public occasions and at inter-collegiate games, but in Western colleges this custom is rarely practiced. Money supplied by over-indulgent parents is the occasion of most of the immoralities. There is no general laxity of college law and sentiment in regard to the morals of the student. Most college authorities deal severely with known cases of drunkenness, theater going, and gambling.

The consensus of opinion among college authorities is that the morals of students are better than those of the same number of youth outside the college. "Our opinion is," says Noah Porter, "and we believe it will be confirmed by the most extended observation and the most accurate statistics, that there is no community in which the pre-eminently critical period of life can be spent with greater safety than it can in the college." President Timothy Dwight bears this testimony: "There is no community of the same number anywhere in the world which has a better spirit, or is more free from what is unworthy, than the community gathered within our university borders. The religious life of the community has been earnest and sincere. The proportion of Christian men in the university is very large, and the influence exerted by them is manifest in its results."

President Thwing says: "I do believe, and believe upon evidence, that the morals of the American college student are cleaner than the morals of the young man in the office, or behind the counter, or at the bench. His life and associations belong to the realm of the intellect, not to the realm of the appetite. His discipline is a training in that virtue the most comprehensive of all virtues—the virtue of self-control. He is able to trace more carefully than most the relations of cause and effect in the sphere of moral action. He recognizes the penalties of base indulgence. It is, therefore, my conviction that the college man is at once less tempted to the satisfaction of evil appetites, and less indulgent towards this satisfaction, than are most young men."

The *expenses* in college vary according to the means and dispositions of the students themselves. In making general estimates, it is impossible to be strictly accurate.

The average cost per year of an education at Harvard is estimated at about \$900; at Yale and Columbia, \$700; at Princeton, Boston, Cornell, and Amherst, \$600; at Wellesley, Smith, and Vassar Colleges, \$500 to \$600. The average cost of an education in most Western colleges does not exceed \$300 or \$400. At Oberlin College, Wooster University, and the Ohio Wesleyan University the average yearly expenses are reduced to \$200 or \$250.

It is evident that higher education is more expensive in Eastern than in Western colleges. The difference arises from various causes. The tuition ranges from \$100 to \$150 in Eastern colleges, and from \$30 to \$50 in Western colleges. Again, the professors in most of the Western colleges receive smaller salaries than those in the Eastern colleges. In many of the smaller college towns the cost of living is low.

Then the student's personal and social habits play an important part in making up the general average. The large room rent and elaborate furnishings, expensive athletic sports, and costly fraternity life is much more manifest in the Eastern than in the Western colleges. The students are prone to follow the standards of home expenses, and fall in with the spirit of the wealthy social class, and indulge in elaborate living. Parents should discourage any display of wealth or extravagance in college if they wish their sons not to spend their time attending clubs, theaters, and questionable places of amusement, but to devote their attention to attaining true scholarship.

The student's manner of living varies according to location and circumstances. In Eastern colleges the students reside mostly in dormitories located on the college campus, or in fraternity chapter houses, and secure their board outside in clubs or restaurants. These rooms rent from \$50 to \$300 a year, and the price of board varies from \$3 to \$7 per week. The dormitory system does not prevail to any great extent among Western colleges. Students rent rooms in private residences, paying from 50 cents to \$2 per week, and find board in families or clubs at a cost of \$2 to \$3 per week. The students boarding in clubs are comparatively free from restraints, and often fail to cultivate the social amenities and table manners which should characterize a cultivated gentleman. For this reason, boarding in private families, where a woman's presence usually lends grace and dignity to social life at the table, is better for the student. The college student cannot afford, for the sake of cheapness in club life, to become rude or coarse. The people look to the college-trained man for that inherent polish which reveals itself in good taste and refined manners. Success and usefulness in life often depend upon these small matters.

The students in American colleges are not measured by social and financial standards. The colleges sustain democratic ideals and methods by discouraging costly luxury, and encouraging simplicity of living without

making life bare of all that is elevating and refining. They believe that "plain living and high thinking" is the way to call out the talent hedged about by financial difficulties, as well as to spur those gifted with fortune to higher aims and nobler efforts. The student who has the promise of a large inheritance has intimate social relations with those whose only capital is brain and heart. The true college test is thus expressed by President Thwing: "Brain is the only symbol of aristocracy, and the examination room the only field of honor; the intellectual, ethical, spiritual powers the only test of merit; a mighty individuality the only demand made of each, and a noble enlargement of a noble personality the only ideal." This is a healthful condition in college life, and tends to develop in the student self-respect and independence as an essential element in true citizenship.

Students of limited means are encouraged to secure an education. The young man of ability and perseverance, who commands the esteem of the college community, will receive encouragement and support to complete his course in college. There are many charitable foundations to help a needy young man in college. Harvard gives away annually to students nearly \$100,000 in prizes, scholarships, and fellowships. Cornell has six hundred free scholarships, and other colleges deal generously with earnest and worthy students. The hesitating young man who desires an education would do well to follow Franklin's advice, "Young man, empty your purse in your head." If necessity requires that the student should go through college poorly dressed and with plain living, he can afford to face these apparent disadvantages when he is confident that within a few years, by force of application, he can win a position of honor and independence as the reward of true merit. It is a significant fact that the majority of the students in our American colleges come from homes of moderate means, and that fully one-third are earning their way through college.

VI.

THE PERSONAL FACTORS IN A COLLEGE EDUCATION.

One of the personal elements entering into a college education is the choice of a college to attend. This decision is a problem of the first importance, and should not be left to ignorance or caprice, but ought to be carefully considered, inasmuch as it largely involves the future type of character a student will have after the formative period of college life. The college puts a life-long stamp upon its graduates. It largely shapes their tastes, determines the company they keep, and greatly influences the serious work of their lives. There are a few principles by which we may test the excellence of a college without undue disparagement of any.

In the first place, a young man or woman should select a college where the standard of scholarship is high. The number and extent of studies in the college curriculum is not so important as the quality and tone of instruction. The world has come to require accuracy and thoroughness in instruction. What little a student knows he ought to know thoroughly, and then he can speak and act with assurance. A low intellectual tone or lack of critical work on the part of a college has a debilitating influence on the student. The professors should have a ripe scholarship, and be earnest and strong in their work, as well as inspire scholarly ambitions. Their bearing should be kind, courteous, and gentlemanly, in order that the students may come to possess more manly and womanly qualities of character as well as scholarship. Such teachers, in close personal contact with students, will awaken new powers, and help to discipline the mind to clear thinking, and impart noble impulses that will enrich manhood and womanhood.

Again, the college buildings, libraries, apparatus, and general equipment are important, but not as essential as the teaching force. President Gates says: "Harvard ranked as a small training college, and had no cabinets illustrative of science, when she trained Emerson and Holmes and Lowell, among all her gifted sons still her triple crown of glory. Bowdoin had no expensive buildings upon her modest campus when Hawthorne and Longfellow there drank at the celestial fount. Amherst, among her purple hills, boasted no wealth of appliances or endowment when she printed the roll of undergraduates rendered forever illustrious by the names of Richard S. Storrs, Henry Ward Beecher, and Roswell D. Hitchcock. Presidents Woolsey and

Wayland, and Mark Hopkins and Martin B. Anderson, were trained for their noble and ennobling work in colleges which lacked rich appliances and thronging numbers." Such, however, has been the growth of the sciences and advancement in the methods of teaching, that in our modern schools for superior instruction the well-equipped college has a decided advantage over those with meager appliances.

Likewise, select a college where the life and *esprit de corps* is the very best. The college is not an exercising ground for the intellect alone, but a place for inspiring ideas and aims. These are the soul of college life. They are more important than college buildings, endowment or libraries.

The religious principle should have the ascendancy in the choice of a college, because religion demands the supreme place in life. The moral and religious character is by no means fixed when the student enters college, and he needs to come into a pure Christian atmosphere, where the heart, as well as the mind, is molded and stimulated.

Other things being equal, the student should favor a college of his own denomination, or the one that he thinks best represents the spirit and form of Christianity. His church affiliations should be strengthened. In advising this, we do so not from any sectarian bigotry. The probabilities are that if the student attends a college of another denomination, the impressions made may tend to produce indifference to the church of his fathers, or weaken his own Christian efficiency in it. The young should maintain personal loyalty to the church that has helped to build up their Christian character and to inspire in them a thirst for a broader culture.

It is claimed to be an advantage to the student living in the West to select a college in his own state, where he will form his friendships and associations, which afterward may be of value to him in his chosen profession. In such cases, it is thought advisable to take graduate work in the East, in some university which is pre-eminent for its special courses, libraries, laboratories, and appliances. On the other hand, it would often be an advantage for the Eastern student to take work in the best universities of the West.

We come now to speak of some of the *personal hindrances and advantages* in acquiring an education. Student life has its hindrances. All have not the same capacity to assimilate culture. It requires more effort for some to master a college course than for others. A thorough college training costs arduous labor. Many are not willing to pay the price, and to practice the self-denial necessary to acquire the power to think and master the great subjects of study. It demands all the force of a strong conviction and an earnest resolution to go through college and win a place among the thinkers of the world. One reason why so many students enter college and drop out before they complete their course of study, arises from the fact that they have not acquired the power of application. Their feeble wills and intellectual lethargy succumb before mental tasks requiring eight or ten hours of hard, earnest work a day. They should be encouraged with the words of Lord Bacon, who says: "There is no comparison between that which we may lose by not trying and not succeeding, since by not trying we throw away the chance of an immense good, and by not succeeding we only incur the loss of a little human labor."

Again, there are those who are led to look for some short cut to obtain a college education. This is a serious mistake. "Whatsoever a man soweth, that shall he also reap," is as true in an intellectual career as in any other work of life. The laws of mental growth must be observed to make the most of ourselves, and to do the most for humanity and God. The young must learn that it takes years of work to get a college education. "If I am asked," says President J. W. Bashford, "why Methodism does not produce more John Wesleys, I assign as one reason of this failure the fact that none of us observe the laws of mental development as John Wesley kept them, and devote the time to mental growth which John Wesley gladly gave. I turn to Arminius, and find that he spent between twelve and thirteen years at the universities of Europe before he began to preach. Arminius died at fifty-nine. Yet he left behind him a work on divinity which ranks him with La Place and Newton, with Calvin and Augustine and Spinoza, as one of the world's master minds. Calvin spent nine years at college, and later was able to devote three years more to study. Augustine devoted thirteen years to study after his father sent him away to college before he accepted the professorship at Milan. It was eleven years after Luther left home for college before he left the scholar's bench for the professor's chair. Four years later, Luther took another scholastic degree, showing that he was still pursuing his studies. Five years more were required for

Luther to reach clear convictions on religion and theology. Paul was a student in the most celebrated schools in Jerusalem for fifteen years. If, therefore, you do not seem to have that mastery of truth, if you do not find yourself the intellectual giant which you once thought you might become, do not blame the Lord, do not depreciate your talent, until you have devoted as many years to college studies as did Arminius, and Calvin, and Augustine, and Wesley, and Luther, and Paul. If you would do a great work in the world, fulfill the conditions by which men outgrow their fellows." The student should be willing to begin at the bottom of the ladder and work upward. It will take more time, but it will yield rich returns and bring real satisfaction.

Again, if the college life is to be profitable and pleasant, the student should refuse to enter an advanced class when his general culture or discipline is so deficient as to render it difficult to make reasonable progress in his studies. It is true that the entrance examination is not always a fair test of the student's capacity or promise. The difficulty cannot be corrected, and study be made a pleasure, unless a student himself shows frankness, and is willing to begin where every step forward is thoroughly understood.

Among the *personal advantages* of a college education is the fact that it helps to *emancipate the individual*. The studies pursued take the student out of his narrow self and his present environment, and make him conversant with other ages and conditions, where he finds his larger self. The personality becomes enlarged and enriched by a wider vision and a knowledge of the great and good men who have lived to make the world better. The best thoughts of the past and the present are at the student's command. He may place himself in touch with all ages and peoples and feel that he is contemporaneous with the best spirit and thought of all that have gone before. Truth thus gathered and stored up in life and character has a wonderful emancipating power. The gateway of truth is always thrown open to those who earnestly knock and search for her hidden treasures. The individual in this age, more than in any other, needs the emancipating power of truth to act intelligently and effectively in the drama of life.

A college education likewise *tends to liberalize the individual* by first eliminating any self-conceit, or inclination to rashness or falsity, and to build up firmness, judgment, and sincerity of character. The aim of the college is to enable the student to know himself and his mission in life. He must have a right conception of self, because he must everywhere live and act with self. He owes it to himself, and to the race, and to God, to make the most of life by developing his God-given faculties. God had a purpose in creating each person, and the aim of each individual should be to live worthy of his origin, by finding out what God wants of him, and then training his faculties and aptitudes on the line of this purpose. He who lives in willful ignorance lives beneath the privileges and possibilities of a human being created in the divine image. No one ought to be satisfied with anything short of the noblest and best possibilities for himself. The majority of men and women have rich capacities, and their natures are full of resources, but these are not always called out. Their incipient powers often need some outside impulse or suggestion to open the chambers of the soul and lead them to discover their unconscious capacities, natural aptitudes, and untried powers.

There are hidden forces in our nature and in life about us of which we little dream. The marvelous forces of electricity are being applied to all human activities, and are unfolding to us new life and new possibilities. We are told that there are mightier currents in the atmosphere above us than those in the Mississippi or the Amazon. Likewise, the science of education exhibits how the trained powers of man reveal unexpected forces and capacities, which have needed only the touch of truth and personality to awaken a higher life and to impart fresh inspiration. Now the college is the best place to discover our inborn energies, and to awaken talent and develop greatness through the influence of men and books.

The student is also liberalized by a knowledge of the truth. Ignorance is the synonym for narrowness and bigotry. Charity, good-will, and human brotherhood spring from a kind heart and an enlightened understanding. The student, by reason of years of study, is better able to see truth in its various human relations and personally exhibit a breadth of charity unknown to those of narrow vision. His informed judgment and quickened conscience will enable him to act generously and to suffer courageously, because his soul is quietly resting in the bosom of truth.

A college education likewise *helps to fortify the individual* for complete living. It is in the college that the student gains a deeper consciousness of his own ability, which gives independence to character. Through genius, or by dint of extraordinary application, he attains an intellectual ability which gives him the right to wield his trained powers to uphold the truth and work for the general good. His mental powers, stores of knowledge, and humanitarian sympathies naturally give greater opportunity for influence and usefulness. The judgment and reasoning powers have been trained so that the student goes forth fortified against the acceptance of plausible delusions and sophisms, and can speak with rightful authority as to the facts or principles he has observed and verified. Truth and personality, thus coupled together, face practical duties and questions with the confident strength and heroic courage which presage victory.

The college-trained man, who enters his vocation in life as a vigorous, virtuous and capable being, equipped with facts and principles as the propelling power of life, will wield the greatest influence for good. He will be fortified for the battles of life, and able to maintain himself in honest independence.

The college offers another safeguard to the student by conserving scholarly tastes and habits. The student who acquires a literary taste is never at a loss to know how he may best employ his time. The baser things of life are crowded out to give place to nobler thoughts and higher aims. He finds his real happiness in cultivating the inner life of exalted thought and generous impulses. He realizes that, as the body demands sustenance, and the soul needs "bread from heaven," so the mind must have intellectual food, which gratifies a taste for the best thoughts of the best thinkers.

The student is also helped to fortify himself with a noble purpose. He is led to feel that he has a mission in life, and the power of this purpose gives an elevation to the spirit and a dignity and loftiness to conduct. More than anything else, it helps to strengthen the will to resist temptation and to conform to the highest moral code. By far too many of our youth are drifting through life without any particular aim or purpose. They fail to act in life under the inspiration of a devotion to a great purpose. Henry D. Thoreau was right when he wrote: "The fact is, you have got to take the world on your shoulders, like Atlas, and put along with it. You will do this for an idea's sake, and your success will be in proportion to your devotion to ideas. It may make your back ache occasionally, but you will have the satisfaction of hanging it or twirling it to suit yourself. Cowards suffer; heroes enjoy." Any worthy calling or useful employment will lead to honor and a broader development of self, providing that self is filled with an absorbing love to God, so that it will be the unit of measure for action towards a neighbor and the true base line from which his rights and boundaries are surveyed and determined.

The college helps to fortify the young by imparting good impulses, which enable them to enter upon life full of hope and courage. It is the place to kindle the youth with a glow of enthusiasm, and impart an inspiration which will pervade the whole career of life. It speaks for the immaterial and unseen forces of life, and supplies the purest motives by which to form a true and beautiful character.

No young man can afford to enter the wide-open door of the twentieth century without a harmonious development of his faculties, and a nature sensitive to the best and holiest influences, and responsive to the most generous impulses. The aspirations of bright minds and noble natures can never excel the lofty descriptions of wisdom by the wisest of men.

"Happy is the man that findeth wisdom
And the man that getteth understanding,
For the merchandise of it is better than silver,
And the gain thereof than fine gold.
She is more precious than rubies,
And all things thou canst desire are not to be compared unto her.
Length of days is in her right hand,
And in her left hand riches and honor;
Her ways are ways of pleasantness,
And all her paths are peace."

VII.

THE PRACTICAL VALUE OF EDUCATION.

Prince Bismarck is reported to have said that in Germany "there were ten times as many people educated for the higher walks as there were places to fill." Many uninformed persons are ready to make similar statements in regard to this country, and believe that we are over-educating the people. Colonel R. G. Ingersoll says: "You have no idea how many men education spoils. Colleges are institutions where brickbats are polished and diamonds dimmed."

The public schools have nearly fifteen million pupils enrolled, or nearly one-fourth of the population of the entire country. In 1890, the four hundred and fifteen colleges had 118,581 students in all departments. This vast army of youth receiving instruction is regarded, on the part of some people, with a little disquietude, and it is believed that we are likely to have too many college-trained men and women. There are certainly no grounds for fear if we take education to mean the broadest culture for complete living.

If we examine more closely the figures regarding our school population, we will find that, of the large number of pupils enrolled in 1890, there was only "an average of three and one-half in one hundred pupils studying any branches above the courses of study laid down for the first eight years; that is, between the ages of six and fourteen years."

Of the 118,581 students in our colleges, there were only 35,791 men and 7,847 women in the collegiate department, making a total of 43,638 receiving higher instruction. The remaining number were in the preparatory, normal, and professional departments. These students are scattered over a great nation, and if we take students in all departments they represent one in four hundred and fifty-five of the population who are under superior instruction, and only one male student in the collegiate department to a group of 1,770 of the population. Many of those enrolled in college do not complete the course of study. It is evident that the number of students in our colleges is proportionately small, considering our population and the requirements of our age, and the proportion of graduates is even smaller.

The practical value of a college education is seriously questioned by many good people unacquainted with the facts. There is abundant evidence, however, which goes to prove that the college graduate has better chances for success than the non-graduate.

It is admitted at the outset that some self-educated men have succeeded without a college education, while some college-trained men have failed in active life. It should be remembered that colleges do not exist to make ability, but to develop it. There is certainly nothing in a college education which unfits men for the practical duties of life. Some college students have meager talent to begin with, and a college training aims to help them make the most of themselves.

The so-called "self-made" men have undergone the severest discipline. By force of native ability and energy, they have surmounted difficulties and achieved success which merits the warmest praise. There is scarcely one of them who would not have availed himself of a collegiate or technical training if force of circumstances had not ordered otherwise. They feel keenly their educational disadvantages, and believe that they would have had greater success if they could have had the disciplinary training of a college course. Many feel as did the distinguished orator, Henry Clay, who, when in Congressional debate with John Randolph, a collegian, is said to have acknowledged, with tears, the disadvantage he suffered from not having had a liberal education.

Washington, Franklin, and Lincoln achieved success by their application, but they were among the foremost to recognize the value of a college training. These examples show that a college education is not always essential to the highest service. The only just claim for a collegiate training is that it increases the probabilities of a person's success in life.

The criteria of comparison of the achievements of men are imperfect, and the measure of success is not easily calculated. Great men are not those who simply climb up to some conspicuous position. It is important to estimate the quality of the work done, as well as the place occupied. A greater premium should be placed upon the manhood and womanhood put into the work, rather than the place filled. The teachings of Christ show that there is no place in the Kingdom of God for a place hunter, but that greatness is measured by service. In the competition for success in life, it is often necessary to have not only ability and worth, but the commercial instinct to gain public recognition. The safe rule for men of talent to follow is to make themselves conspicuously great in their present position, and make it a stepping-stone for something greater. Charles Kingsley occupied, in England, an apparently humble position in his rural pastorate, but the thinking world has felt the power and influence of his great life.

Bearing in mind these restrictions in regard to the idea of success, we offer a few suggestive facts to show the number of college men who have made a record in the annals of the country.

The college has been the open doorway to positions of eminence and usefulness in all countries. Lord Macaulay, in one of his speeches in Parliament, said: "Take the Cambridge Calendar, or take the Oxford Calendar for two hundred years; look at the church, the parliament, or the bar, and it has always been the case that men who were first in the competition of the schools have been first in the competition of life."

Speaking of the advantages of a university education in Germany, Professor J. M. Hart says: "I am warranted in saying that the majority of the members of every legislative body in Germany, and three-fourths of the higher office holders, and all the heads of departments, are university graduates, or have at least taken a partial course—enough to catch the university spirit. All the controlling elements of German national life, therefore, have been trained to sympathize with the freedom, intellectual and individual, which is the characteristic of the university method."

It is estimated that only one-half of one per cent. of the male population in America receives a college education, and yet this small contingent of college men furnishes one-half of the Senators and Vice-Presidents, two-thirds of the Presidents and Secretaries of State, and seven-eighths of the Justices of the Supreme Court of the United States.

Rev. W. F. Crafts says: "I have examined the educational records of the seventy foremost men in American politics—Cabinet officers, Senators, Congressmen, and Governors of national reputation—and I find that thirty-seven of them are college graduates; that five more had a part of the college course, but did not graduate, while only twenty-eight did not go to college at all. As not more than one young man in five hundred goes to college, and as this one-five-hundredth of the young men furnishes four-sevenths of our distinguished public officers, it appears that a collegian has seven hundred and fifty times as many chances of being an eminent Governor or Congressman as other young men."

The college graduate generally has the pre-eminence among professional men. The proportion of successful men in the professions is difficult to obtain, but if a wide reputation be regarded as the criterion of success, the college-bred men take the lead.

President Thwing has carefully estimated that, of the 15,142 most conspicuous persons of our American history, whose record is sketched in "Appleton's Cyclopædia of American Biography," 5,326 are college men. Among the latter, the percentage found in the various callings is as follows: "Pioneers and explorers, 3.6 per cent.; artists, 10.4 per cent.; inventors, 11 per cent.; philanthropists, 16 per cent.; business men, 17 per cent.; public men, 18 per cent.; statesmen, 33 per cent.; authors, 37 per cent.; physicians, 46 per cent.; lawyers, 50 per cent.; clergymen, 58 per cent.; educators, 61 per cent.; scientists, 63 per cent." He further estimates that one college man in every forty attains recognition, to one in every ten thousand non-college men; and a college-bred man has 250 times the chance of attaining recognition that the non-college man has.

Dr. Channing says: "The grounds of a man's culture lie in his nature, and not in his calling;" and, in keeping with this, the primary aim of a college is to train men. Yet, it should be the door of approach to all professions. The studies pursued in college are the foundations of the practice of the various professions, and a young man

does himself and his profession no credit when he neglects to master a college course because of his impatience to rush into a professional career, and thus help to swell the army of poorly-equipped professional men.

"To practice law or medicine in France," says Matthew Arnold, "a person must possess a diploma, which serves as a guarantee to the public that such a person is qualified for his profession. A licentiate of law must first have got the degree of Bachelor of Letters; have then attended two years' lectures in a faculty of law, and undergone two examinations, one in Justinian's Code, and the Codes of Civil Procedure and Criminal Instruction. The new bachelor must then, in order to become licentiate, follow a third year's lectures in a faculty of law; undergo two more examinations, the first on the Institutes of Justinian again, the second on the Code Napoleon, the Code of Commerce, and Administrative Law, and must support a thesis on questions of Roman and French Law. To be a physician or surgeon in France, a man must have a diploma of a doctor either in medicine or in surgery. To obtain this, he must have attended four years' lectures in a faculty of medicine, and have two years' practice in a hospital. When he presents himself for the first year's lectures, he must produce a diploma of Bachelor of Letters; when for the third, that of a Bachelor of Sciences, a certain portion of the mathematics generally required for a third degree being, in his case, cut away. He must pass eight examinations, and at the end of his course he must support a thesis before his faculty."

Young men with talent and ambition are led to believe that the professions are so over-crowded that there is very little opportunity, in these days, for a collegian to succeed in a professional career. A comparative study of the number of students in the professional schools in Germany, France, and the United States, for 1890 reveals the following facts:

	<i>Law.</i>	<i>No. to every 100,000 population.</i>	<i>Medicine.</i>	<i>No. to every 100,000 population.</i>	<i>Theology.</i>	<i>No. to every 100,000 population.</i>
Germany,	6,304	13	8,886	18	5,849	12
France,	5,152	14	6,455	17	101	..
United States,	4,518	7	14,884	24	7,013	11

We glance briefly at the promises which the so-called learned professions hold out to young men. The opening for young men in the legal profession has many difficulties, but it is not without its rewards. David Dudley Field estimated that in 1893 there were 70,000 lawyers in the United States. If we estimate the population of the nation at 70,000,000, there would be one lawyer for every 1,000 of the population. Assuming that three-fourths of the population are women, children, and men under age, there would be one lawyer to every 250 males of full age in the United States.

Germany, with a population of 50,000,000, has about 7,000 lawyers, or one to every 7,000 persons. In the State of New York, with a population of 6,000,000, there are 11,000 lawyers, or one for every 545 of the population. Of this number of lawyers, there is a great proportion engaged in real estate business, or other outside matters, which enables them to secure a maintenance. Others have entered the law because of its promise of social position and honor.

Aside from the numbers in the legal profession, there are other considerations in the problem. The people of to-day are less disposed to controversy, and avoid employing lawyers to settle disputes and differences in court, and others often hesitate to employ a lawyer for fear of being made a victim of the rapacity of some who have brought the profession into disrepute. Again, there is less confusion in the laws. They are being collected, condensed, arranged, and simplified, and people are coming to understand the codes. Likewise, the courts are adopting simpler rules and codes of civil procedure, which give less room for pettyfogging hindrances and delays in litigation. A lawyer of talent, with the aid of a good stenographer and typewriter and other advantages of to-day, can do double and treble the work of a lawyer twenty-five years ago.

Finally, the qualifications of a lawyer never reached so high a standard. To attain the greatest professional success, it is indispensable to get the highest development which a college training can give. Chauncey M. Depew says that three-fifths of the lawyers are unfit for their profession from lack of ability or training. The people demand abler and better lawyers. The requisite qualities of a good lawyer to-day are not only knowledge and a good judgment, but patience, industry, honesty, and certain other aptitudes for his work. He must be ready to compete with a trained and talented rival. Special training is of great value. A lawyer of several years' standing at the bar in New York, in a recent conversation, remarked: "I studied law in a lawyer's office. My brother, here, several years younger than myself, went through the law school, and he has so much the advantage of me, in consequence of that training, in the studious habits he has formed, in being brought into immediate contact with the best legal minds, in being held to the highest standards, that this fall I shall enter the law school and take the entire course."

In facing these difficulties, let it be remembered that there are always openings for young men of superior qualifications. Some one asked Daniel Webster whether the legal profession was not over-crowded, and he replied that there was always room at the top. An ambitious young man of ability can win his way to the front, while mediocrity will wait for patronage. There is jostling and crowding in the rear ranks of every profession. It is surprising how few thoroughly trained men are entering the profession. In 1890 there were in the various law schools in this country 4,518 students, and only 1,255 of these had degrees in letters or science. In the same year, 1,514 were graduated in the schools of law, which was only 2.4 in every 100,000 of the population. There is a demand for specialists. The field is enlarging in the department of patent law, railroad law, and other legal specialties. The business transactions of this age are more complex, and the interests more important. Corporation controversies need to be adjusted by those who thoroughly understand the principles and practices of equity. "I was a teacher of law to young men for more than twenty years," says Judge Hoadley, "and have never seen any reason to discourage a sober, honest, and industrious young man from studying law. He needs, first of all, absolute fidelity, trustworthiness, and integrity; secondly, devotion to his calling—in other words, industry that will not be interfered with by the distraction of society or pursuit of politics. If he be honest and willing to work, he will, with reasonable intelligence make a sufficient success, if he have the patience to wait for success. If, in addition, he have what I may call the lawyer's faculty—that God-given power to appreciate leading principles and apply them to facts as they arise, coupled with ability to reason, and to state results cogently and persuasively,—he will make a shining success."

Again, the advantages of a thorough medical education are generally recognized. The sacred work of ministering to the suffering demands the most thorough instruction in medicine and methods of treatment. In 1890 there were 15,404 students in 116 medical schools in the United States, distributed as follows: Regulars, 13,521; eclectics, 719; homeopaths, 1,164. For the same year there were 4,492 graduates, or 7 in every 100,000 of the population. Sixteen of the medical schools had no students enrolled who had previously obtained a literary or scientific degree. Only 15 per cent. of all the students matriculated had obtained a degree before entering the medical schools. There is an evident lack of thorough preparation in foundation studies on the part of the students. The medical profession is second to none in importance, and the students of medicine who will give time to the more extended culture of a college course will naturally obtain greater skill and a broader range of thought, which will contribute to their efficiency as practicing physicians.

It is also encouraging to know that the statistics of each decade indicate that an increasing proportion of young men entering the ministry have received a college education. There were 112 theological schools in 1890, that reported 7,013 students, of whom 1,372 were graduated, or two for every one hundred thousand of population. This is certainly not over-crowding.

Of the students in theology enrolled in the schools of the various denominations in 1890, the proportion was as follows: Baptists, 15.6 per cent.; Presbyterians, 15 per cent.; Methodists, 14.9 per cent.; Lutheran, 14.7 per cent.; Roman Catholic, 13.4 per cent.; Congregational, 9.7 per cent.; Christian, 5.5 per cent.; Episcopal, 4.7 per cent.; Hebrew, .5 per cent. Of the total enrollment, 7,013, only 1,559 students had received degrees in letters or science. The church demands educated men for the pulpit. A call to the ministry in these days means that a man should prepare for the work. God does not honor the slothful, but the man who seeks to make full proof of his ministry. This is done when a man of piety takes the time to acquire mental culture and

refinement, and to become able properly to guide and instruct the people. Such ministers, "thoroughly furnished unto every good word and work," honor the church, and strengthen the cause of Christ. Their mental endowments command respect and inspire confidence. There never has been a time in the Christian ministry when there was such a demand as now for ministers with minds cultivated and well stored with knowledge, and hearts set on fire by the Holy Ghost.

The old idea that a college graduate must study for medicine, law, or the pulpit, has attracted a large number of them into these professions. We have learned, however, that these professions are not superior to other avenues in science and business. A college training is only a means to an end. It is giving a man fitness for work of any kind. The departments opening up to college-trained men in all lines of work are multiplying and expanding with each succeeding year.

The future is bright for those who will take up statesmanship as a profession. Nothing has a more important bearing on the social interests of the people than the science of civil government. The nation is burdened with politicians, but intelligent Christian statesmen are few. The intelligent people of this nation are asking for men educated in history, political and social science, who, with clear heads and loyal hearts, will use their ability for the welfare of the public. Good citizens have too long held themselves aloof from the great concerns of our organized society. All civic matters are worthy of our best thought and noblest effort. The management of our political and social interests has too often been usurped by politicians, who, with little self-respect, efficiency, or character, have worked not for the public good, but on the principle that "to the victors belong the spoils." Their rapacity and greed have led them to sacrifice principle to party. They aim to manage caucuses, pervert elections, override the wishes and defy the moral sense of the people, and corrupt the sources of national life.

We have come to ask for a remedy. Its answer must be found in the young men whose patriotism will lead them to thoroughly prepare themselves for public service and make statesmanship a profession. Along with a broad and comprehensive knowledge of the science of government they should cultivate the capacity for effective public speech, in order to present political and social themes with such power as to guide public opinion in the right direction. They must be willing to carry their independent convictions into civil affairs, and help to ennoble the national spirit, and purify public life, and make it expressive of the highest intelligence and the best moral sentiments of the people. Statesmanship is a sacred calling, and the people are ready to uphold and encourage young men who will dedicate themselves to this exalted work.

It is an omen of good that chairs of political and social science are being established in all our high grade colleges to train young men for this service. They ought to prosper, and will. Milton saw this need years ago, and said: "The next remove must be to the study of politics, to know the beginning, end, and reasons of political societies; that they may not, in a dangerous fit of the commonwealth, be such poor, shaken, uncertain reeds, of such a tottering conscience, as many of our great counsellors have lately shown themselves, but steadfast pillars of the state."

Those who are to be trained for this leadership, and expect to gain a strong hold on society, should be taught and trained to think upon complicated questions, and able not only to frame platforms and shape legislation, but to grapple with modern social problems, and lead the people to nobler action.

Journalism is another important field for talented young men and women. The journalists of to-day need breadth and concentration of mind to meet the demands of a reading and thinking people. They need a knowledge based on history, literature, and politics in order to report speeches correctly and to discuss living questions clearly, cogently, and with a broad knowledge of principles and facts. The press wields an influence next to the pulpit, and it should be consecrated to the highest service through men qualified for editorial work.

The profession of teaching has justly assumed a position in this country second to none in influence and power.

There are 15,000,000 pupils in the public schools of this country. There are 364,000 teachers employed in giving instruction to this army of youth. College graduates are rapidly acquiring a control of the high

positions in these schools. The superintendents, principals, and the majority of the male assistants are college graduates. A college education is fast becoming an absolute necessity to secure a position in the best schools. School boards will rarely select a superintendent or a principal of the high school who has not received a collegiate education. There is an increasing demand for thoroughly trained men and women in this work. Few teachers can hope to attain prominence in their profession without these advantages.

There is, likewise, a rich and fruitful field opening up to those who receive a careful scientific education. The application of science to the arts and industries is rapidly changing the social and economic conditions of the people. We are unable to conceive of the ever-widening field in which educated men will be needed to discover new methods of concentrating and transmitting electrical and mechanical power, thereby reducing the cost of production, and adding to the comfort and happiness of the human family. There is a growing demand for men versed in electrical science, who can take charge of establishments for the transmission of power. Civil and mechanical engineers are needed, who can wisely and economically construct our bridges and highways of commerce, and who can apply the highest scientific skill to all the constructive enterprises of the country.

"The Swiss and Germans aver," says Matthew Arnold, "if you question them as to the benefit they have received from their *realschulen* and *polytechnicums*, that in every part of the world their men of business, trained in these schools, are beating the English when they meet on equal terms as to capital, and that where English capital, as so often happens, is superior, the advantage of the Swiss or German in instruction tends more and more to balance this superiority. I was lately saying to one of the first mathematicians in England, who has been a distinguished senior wrangler at Cambridge and a practical mathematician besides, that in one department, at any rate—that of mechanics and engineering,—we seemed, in spite of the absence of special schools, good instruction, and the idea of science, to get on wonderfully well. 'On the contrary,' said he, 'we get on wonderfully ill. Our engineers have no real scientific instruction, and we let them learn their business at our expense by the rule of thumb, but it is a ruinous system of blunder and plunder. A man without a requisite scientific knowledge undertakes to build a difficult bridge; he builds three which tumble down, and so learns how to build a fourth which stands, but somebody pays for the three failures. In France or Switzerland he would not have been suffered to build his first bridge until he had satisfied competent persons that he knew how to build it, because abroad they cannot afford our extravagance.'"

We find, likewise, that our industries are demanding men trained in applied chemistry. The application of the principles of chemical philosophy to manufacturing steel, chemical fertilizers, artificial preparation of articles of food, bleaching, dyeing, and printing of cloths, offers a very inviting field of study. We might multiply instances, but enough has been said to suggest to our minds the rich possibilities before educated young men and women. We are only on the edge of the future of applied science.

We need, also, to carry our culture and training into business careers. Business is conducted by different methods than in the past. The management affords a broader field for judgment and thought. Many, in the future, may succeed without a college education, but they will work at a disadvantage. The chances are always in favor of the man who is well educated. It is a common belief that a college education unfits a man for practical work. He often does appear at a disadvantage on leaving college, but, other things being equal, he will distance, within a few years, the man of like ability who has not been rigorously trained to see, think, and judge. "Experience also confirms this impression by the decisive testimony gathered from a multitude of witnesses," says Noah Porter, "that the young man who leaves college at twenty-one, and enters a counting or sales-room, will, at twenty-three, if diligent and devoted, have outstripped in business capacity the companion who entered the same position at sixteen and has remained in it continuously, while in his general resources of intellect and culture he will be greatly his superior."

Germany has for more than fifty years insisted that her youth should not only have the foundation of a general education, but that opportunities should be given for higher commercial instruction. This superior education and training is producing its legitimate results. Notwithstanding the many unfavorable circumstances which have combined to prevent her growth in commerce and industry, Germany has gained an amount of skill and

experience in mercantile training that has no parallel in France, England, or America. The advance of German trade is due to the superior fitness of the Germans through their systematic training in technical schools.

M. Ricard, in his report to the French Chamber of Commerce, said: "Every intelligent man must admit that the invasion of our commerce by foreigners is due entirely to this educational inferiority. The Germans are taking our places everywhere. They even supplant the English. Let the merchants of France take warning in time. German commerce has better instruction, better discipline, and greater enterprise than French commerce; it is at home everywhere; no languages are foreign to it; it keeps a lookout over the world; it is not ashamed to go to school, and if you do not awake from your lethargy, it will annihilate you."

The London Chamber of Commerce found, on examination, that ninety-nine per cent. of Englishmen who take to commercial life are unable to correspond in any foreign language. The comparative disadvantage, on all commercial lines, of England with Germany, is owing to "a higher average of mercantile intelligence all round." It is not to be alleged that the English are mentally inferior to the Germans, but, as Professor W. G. Blackie said before the Educational Institute of Scotland: "The question is solely an intellectual one, and must be solved through educational means. It assumes the aspect of an educational duel between the mercantile population of this country and their competitors on the continent, in which the mastery is sure to remain with those who are the most fully equipped for the contest."

The report on the superior instruction of Antwerp contains the following words: "Men have seemed to imagine that, in order to prosper, commerce and industry have only required money and favorable treaties of commerce. Governments have occupied themselves with the material side of the future merchant, without taking care to develop his intellectual capacity, which is, indeed, the spirit of his operations, without taking care to improve his intelligence, which is the germ of enterprise in the commercial life of a nation."

Young men and women are often led to believe that there is no chance for them to have a successful career, and so fail to attend college and develop their capacity, and, as a consequence, often become restless and idle. But this is no age for triflers. The world is in need of educated men in all of the higher walks of life. There is abundant room for men of ability and culture who can bring things to pass. The fact that earnest, talented, and consecrated men and women are overworked in their professions shows that there is a place in the front ranks of all useful professions and vocations.

The door of the twentieth century swings open and invites the ambitious men and women of talent and consecration to the service of humanity, and extends the widest opportunities and the most exalted privileges ever vouchsafed to man. Will the youth of the land be ready to enter?

VIII.

OUR INDEBTEDNESS TO COLLEGES.

The American colleges hold the most intimate relation to the whole community, for which they have done a vast work. They rightly enjoy the confidence and esteem of the American people, since they have infused into society some of the most purifying and life-giving influences. Many of the first settlers were among the best educated men of England, and they recognized that education was the corner-stone of civil and religious liberty. Pembroke, Delaware, William Penn, Roger Williams, the Winthrops, and a large number of worthy men who settled in the early colonies came from the classical shades of Oxford and Cambridge, and retained the educational predilections which were so firmly established in their mother country. The spirit and principles of our wise and godly ancestry were early introduced into the colleges, which have conserved and perpetuated them down to the present day.

The American people owe much to the colleges for training capable and worthy men to fill the posts of honor and power in the nation. The men who have given shape and character to the early political organizations and spirit have been mostly collegians.

These institutions for higher education have trained men in history, philosophy, and the principles of government, who have become the right hand of strength to the nation. Their extensive knowledge and thoroughly disciplined and comprehensive minds have been largely instrumental in perfecting our system of government, and in elevating the nation to the rank of one of the greatest political powers.

The colleges have trained the intellect and conscience of the majority of students so that they have gone forth as leaders, and have exerted a prodigious influence among the people for right thinking and right acting. They have not only disciplined the powers of the masterly statesmen, but have fostered among them a sense of fraternity concerning our civil destinies. The students that have been gathered into the colleges from the different portions of the nation have become imbued with one sentiment, and entered upon public life linked together by the bonds of a common intellectual life and strong friendships, which have resulted favorably for the republic.

Some of the colonial colleges have richly repaid the nation for all the effort and sacrifice it cost to found them. William and Mary College has sent out twenty or more members of Congress, fifteen United States Senators, seventeen Governors, thirty-seven Judges, a Lieutenant General and other high officers of the Army, two Commodores to the Navy, twelve professors, seven Cabinet officers; the chief draughtsman and author of the Constitution, Edmund Randolph; the most eminent of the Chief Justices, John Marshall, and three Presidents of the United States.

Harvard has furnished two Presidents, one Vice President, fifteen Cabinet officers, twenty Foreign Ministers, twenty-nine United States Senators, one hundred and four Congressmen, and nineteen Governors.

Princeton has beaten the Harvard record in everything except the first and fourth items. It has given to the country one President, two Vice Presidents, nineteen Cabinet officers, nineteen Foreign Ministers, fifty-five United States Senators, one hundred and forty-two Congressmen, and thirty-five Governors.

The collegians have ranked among the principal leaders in the political life of the nation. Fifty-eight per cent. of the chief national offices have been filled by them. Thomas Jefferson, author of the "Declaration of Independence," was a college man. Hamilton, Madison, and Jay, who took such a prominent part in the framing of the Constitution of the United States, were college-trained men. Three-fourths of the signers of the Declaration of Independence were college graduates. These and other superior men in public life, at this period, were educated and possessed a scholarship that was in compass and variety more than abreast with the learning of the time. George Washington was a self-made man, but he had recourse to America's greatest statesman, Alexander Hamilton, a graduate of Columbia College, in preparing his state papers.

The counsellors of Abraham Lincoln, during the stormy days of the Rebellion, were men of trained minds. "All the leaders," says Professor S. N. Fellow, "in that Cabinet were college-trained men. William H. Seward, the shrewdest diplomatist, who held other nations at bay until the Rebellion was throttled; Salmon P. Chase, whose fertile brain developed a financial system by which our nation was saved from national bankruptcy, and made national bonds as good as the gold in foreign markets; Edwin M. Stanton, that man of iron, who organized a million of raw recruits into an army equal to any in the world; Gideon Welles, who, almost from nothing, created a navy sufficient for our needs,—each of these, and every other member of Lincoln's Cabinet, save one, was a college graduate. So, also, in the army. It was not until thoroughly trained and disciplined men filled the chief places in command that the Federal forces overwhelmed and destroyed the Rebellion. We repeat, the law is, and it is believed to be universal, that the higher the rank or position, the larger per cent. of college graduates are found in it."

Education was an important factor in deciding the issues of our Civil War. Thoroughly trained and disciplined men filled the chief places in command in the Federal Army. The Northern soldiers were better educated than those of the South. It has been said that "in the German Army that fought the battles of the Franco-Prussian

war, those who could neither read nor write amounted to only 3.8 per cent., while in the French Army the number amounted to 30.4 per cent." According to the admission of the defeated, the universities conquered at Sedan. Perhaps it is not too much to say that the great number of colleges in the Northern States conquered at Appomattox.

A large per cent. of the leaders in the American Congress, during the trying period of our country's history from 1860 to 1870, were either college graduates or had taken a partial course in college and gained its inspiration.

The college graduates have furnished 33 per cent. of the Congressmen, 46 per cent. of the Senators, 50 per cent. of the Vice Presidents, 65 per cent. of the Presidents, 73 per cent. of the Associate Judges, and 83 per cent. of the Chief Justices of the Supreme Court of the United States.

Again, we are especially indebted to the colleges for encouraging private and public schools, through which we have become an enlightened people. It is impossible to estimate the indebtedness of popular to collegiate education. There is an intimate and vital relation between the college and the public schools, which differ not in kind, but only in the degree of instruction. "The success and usefulness of common schools," says Professor W. S. Tyler, "is exactly proportioned to the popularity and prosperity of the colleges, and whatever is done for or against the one is sure to react, with equal force and similar results, upon the other."

The colleges have been foremost in advocating that the education of the youth should not be left to those of meager attainments and narrow sympathies. They have maintained that, in order to reap the best advantages of our public schools, it is important to have wise, competent, Christian men and women to give instruction, as well as to prepare text-books, and to increase the appliances employed in teaching.

It has been a difficult task to bring our public school system to the present condition of progress. The work has proceeded slowly and steadily under the example and inspiration of great educational centers. The excellence and usefulness of our school system has advanced just in proportion to the culture and ability of the teachers. A collegiate education has always tended to foster and encourage higher standards of scholarship among teachers, and this influence has been diffused into the public school system. President Charles W. Sumner truthfully says: "That which leads up to the highest must always be supervised and directed by that which is at the top. A system of elementary and secondary education which does not culminate in the university, and make that the goal towards which its efforts are directed, is an absurdity. There must be good teachers before there can be good schools, and good teachers can only be formed in institutions that are chiefly concerned with knowledge at first hand. This has been a recognized principle in Germany for half a century, or longer; is now almost universally admitted in France, and is the goal toward which the whole civilized world is rapidly moving."

The efficiency of our public schools has been felt in every department of our social organization. They have been a strong bulwark against the influences of a raw and uninstructed foreign population, who, like a tidal wave, have flooded our shores. Some of these have not only been ignorant and infidel, but filled with monarchical ideas and un-American sentiment. The public schools have brought their children into accord with our American institutions, and developed intelligent patriotism. They have taught the youth common rights and privileges, and helped to generate a union of sympathy and sentiment which leads to the consolidation of our society into a homogeneous body.

The colleges, working through the public school teachers, have likewise helped to educate the millions of the unenfranchised and uneducated colored people, and to break up sectionalism, allay party strife, and make for the peace, prosperity, and unity of the nation. Our political safety has called for a wise and vigorous effort to educate the masses and to assimilate the heterogeneous elements into our body politic. The public schools and colleges, with their interdependence, have in a great measure met the demand, and given us a legacy of peace, prosperity, and intelligence enjoyed by all the people.

Likewise, the colleges have contributed largely to the general prosperity and material progress of society. They are the real centers of power of this enterprising and progressive age. "The revival of learning and the

epoch of discovery ushered in the epoch of natural science, which has made possible the epoch of useful inventions."

College-trained men are the most practical and useful of men. They have been the creators of material wealth and prosperity. Their discoveries and inventions have revolutionized business and social life. Every department of life is teeming with the fruits of science and philosophy, which have been largely built up by colleges and college-trained men. Bacon, Newton and Locke were sons of the English universities. Watt and Fulton associated with college men, and "derived from them the principles of science which they applied in the development of the steam engine and steam navigation. Professor Morse, the inventor of the electric telegraph, was not only a college graduate and professor, but made his great experiments within the walls of a university." Likewise, many other scientists, who have demonstrated the limitless possibilities of steam and electricity, and other valuable discoveries and inventions, were either trained in the colleges or received from them the working principles which were essential to their success. These human inventions are of priceless value to the people. The steam engine has contributed greatly to human welfare. It represents, in the United States alone, 20,000,000 horse power in the form of locomotives, or the steam power of 300 horses for each thousand inhabitants. Besides all this, 6,000,000 horse power in stationary steam engines manufacture goods for us. They give the vast force which toils for us, and the laborer furnishes only the guiding power. These inventions have enabled us to increase our wealth at the rate of \$2,000,000,000 a year during the last decade, and helped to make our people sharers in the products of the world, and in all the blessings of civilization.

Professor Huxley was right when he said: "If the nation could purchase a potential Watt, or Davy, or Faraday, at a cost of a hundred thousand pounds down, he would be dirt cheap at that money." Fifty-two of the inventions now prized by the civilized world were made in Germany, and within the influence of her universities. All these discoveries are opening the doors for more wonderful disclosures. All the great industries of the country require men of trained minds and directive intelligence to organize and control them, and the colleges are recognized agencies to help produce them.

Our literature is also largely the fruit of college labor and tastes. The colleges, as centers of intellectual life, have fostered literary tastes in those who have built up and enriched literature. Their libraries and lectures have gathered together men of literary aims and ambitions, so that the seat of the college has become the home of new and grand ideas, which at once encourage literature and science. This congenial intellectual atmosphere has incited many a young person to project noble literary plans.

The majority of great writers have spent years at the university. Lord Bacon outlined his gigantic plan for "the Instauration of the Sciences" during the four years spent in the University of Cambridge. Milton laid the foundations of his classical scholarship in the university. "Newton was matured in academic discipline, a fellow in Trinity College, Cambridge, and a professor of mathematics. He passed fifteen years of his life in the cloisters of a college, and solved the problems of the universe from the turret over Trinity gateway."

The literary influences of our colleges were early manifest in our nation. The scholarship, classical taste, and fine literary style of the superior men in public life led the Earl of Chatham, in the House of Lords, in 1775, to pay "a tribute of eloquent homage to the intellectual force, the symmetry, and the decorum of the state papers recently transmitted from America, which was virtually an announcement that America had become an integral part of the civilized world, and a member of the republic of letters."

The colleges have nourished the conditions out of which a pure, classical literature may grow. Such men as Edward T. Channing, of Harvard, and Webster, Worcester and Goodrich, of Yale, have performed an inestimable service in preparing the way for our mother tongue to be spoken in its purity.

In the line of history, the American colleges have given the nation such men as Bancroft, Parkman, Palfrey, Prescott, Motley, Winthrop and Adams. In the sciences, there are Dana, Gray, Cooke, Walker, Porter, Woolsey and Agassiz. In law and political science, we have Hamilton, Jefferson, Adams, Evarts, Webster, Chase, Choate, Everett and Sumner. These men have been the true architects of the state. The pulpit is represented by such men as Mather, Edwards, Dwight, Storrs, Warren, Beecher, Talmage, Cook, Thomson and Brooks.

Literary genius has been displayed by men like Longfellow, Bryant, Lowell, Holmes, Hawthorne, Mitchell, Holland, Emerson and a host of lights scarcely less brilliant. These men, who have written in a terse and graphic style, received their stimulus and training in college, and are among the bright examples of classical scholarship, and the results of their genius have enriched character and enlightened the world.

The periodical literature reflects the prevailing ideas, sentiments and spirit of the American people. The college-trained men have been especially quick to utilize this throne of power to guide the public mind to right principles and inspiring motives. The colleges must continue to be fountains whence shall flow a pure, earnest, and truthful literature, which will, in a great measure, determine the destiny of the present and future generations.

We are especially indebted to the colleges for the maintenance of the ascendancy of the moral and religious principles which have done so much in unfolding and shaping our national life. The religious sentiment has been the controlling spirit of the nation, and our patriotism has issued from a meditative and religious temper, which the colleges have been foremost in fostering. Nearly all the great religious and reformatory movements have proceeded from the colleges and universities, whereby great good has come to society. "It was through the interchange of students between the Universities of Oxford and Prague that the teachings of Wycliff passed over into Bohemia and issued in the splendid work of Huss. It was from college students of Florence that Colet, and Erasmus, and More caught somewhat of the spirit of Savonarola, and felt the power of truths that emerged in the Italian Renaissance, and made them contribute so grandly to religious liberty in England. It was in the presence of the college students of Germany that Martin Luther nailed his thesis to the doors, and burned the papal bull, and lit the watch-fire of the Reformation that has awaked an answering brightness from ten thousand hills. It was from a little circle of Oxford students that God led forth Wesley and Whitfield to shake the mighty pillars of unbelief in the eighteenth century."

President William F. Warren says: "By means of the great religious movement called Puritanism, the English University of Cambridge shaped, for nearly two hundred years, the intellectual and spiritual life of New England. Emmanuel College, the one in which John Harvard, Thomas Hooker, John Cotton, and many of the early New England leaders were educated, was founded for the express purpose of providing a nursery for the propagation of Puritan principles. Never were the hopes of founders more fruitfully fulfilled. The New World, then just opening, furnished a field of unimagined extent, with motives and social forces and ranges of opportunity which even yet are a marvel. By founding a new England beyond the sea, and planting a new Emmanuel College in a new Cambridge, English Puritanism was enabled to transcend itself, to exchange the attitude of a struggling ecclesiastical party for that of an Established Church. It gained the opportunity to originate a new social order, and to impress itself upon a new age, built upon new and democratic principles. The initial and fundamental covenant out of which grew the chief of all New England colonies—that of Massachusetts Bay—was formulated and signed in ancient Cambridge. In fact, in American Puritanism, with its social, civil, and religious results, may be seen the high-water mark of the intellectual and spiritual influence which, in the whole course of history, have thus far proceeded from the banks of the Cam." The church, in harmony with the genius of Christianity, has always fostered education. It assumes to guard Christianity by directing education as one of its most powerful of organized forces.

The existence and support of colleges are largely due to the Christian Church. They are the offspring of a dominant desire to promote the cause of Christ, and make them powerful agencies for a positive and aggressive Christianity. In the middle ages the pious princes, Charlemagne and Alfred, established schools for the elevation of the clergy. Oxford, Cambridge and Glasgow Universities were established and fostered by the church to educate more fully the clergy. The founders of Harvard College thus described their motive: "Dreading to leave an illiterate ministry to the churches, when our ministers shall lie in the dust." Yale College was founded by preachers for a like purpose. Princeton College was founded "to supply the church with learned and able preachers of the Word." The fact is that prior to the eighteenth century there was no university founded save those established for the glory of God and the good of the church.

The chosen mottoes of the colleges indicate the spirit of the founders. That of Oxford is, "The Lord is My Light;" Harvard, "Christ and the Church;" Yale, "Light and Truth." Eighty-three per cent. of the colleges in

our land were founded by Christian philanthropy, and are under denominational control. The spirit of infidelity does not lead men to make the sacrifices to found colleges. Perhaps there is not more than one in our nation.

The majority of colleges are positively religious. According to Dr. Dorchester, even Harvard, the oldest college in the United States, that wishes to be understood as non-denominational, has been, for more than half a century, "under the direction of a Board of Fellows, all of whom have been Unitarians, except one elected within a few years; and, besides, the theological school of Harvard College is usually mentioned in the Unitarian Year Book as a Unitarian institution." Leland Stanford University is one of the youngest and richest of our American colleges. The regulations declare it to be the duty of the trustees "to prohibit sectarian instruction, but to have taught the immortality of the soul, the existence of an all-wise and benevolent Creator, and that obedience to His laws is the highest duty of man."

Both of these colleges, reported as "non-sectarian," generously provide buildings and pastors for religious services and lectures. Dr. Dorchester believes that one-third of the State universities are under the presidency of evangelical divines. He further states that "in 1830 the students in the denominational colleges were 76.6 per cent. of the whole; in 1884, they were 79.2 per cent."

All the foregoing facts show the strong and enduring progress of Christianity in the United States; that it is "identified with the highest educational culture of the age; that the denominational institutions are incalculably leading in number and students all the undenominational colleges, and that the great principles and blessed experiences of Christianity are voluntarily and intelligently adopted by a far larger proportion of college students than ever before."

The colleges have upheld the vital truths of the gospel by expounding the scriptures, and setting forth their ethical and religious teaching. They recognize that the divine order in saving men is through the inward working of the truth and spirit of God in their souls. Since knowledge is essential to salvation, it is a duty to enlighten men and bring them to understand the divine plan of salvation. The Bible has been communicated to us in foreign languages, and requires prolonged study and extensive knowledge in order that these oracles of God may be known and accepted among men.

The colleges have given a higher efficiency to the Christian ministry. There are those who have obtained their training and knowledge outside of the college who have accomplished great good. There are pious and devoted men who are illiterate, but whose Christian work has been attended with more apparent results than some college-trained ministers. These, however, are the exception. The rule is that those who combine with their piety scholarly acquisitions exert by far the greatest influence for good. The history of Christianity shows how God has raised up a multitude of scholarly men to uphold the supremacy of the gospel over all its foes. Paul, Luther, Melancthon, Calvin, Knox, Cranmer, Wesley and Fletcher were all college-trained men. These men, with others, endowed with mental vigor, great learning and executive force, have been used by God to accomplish His great task of building up His kingdom on earth.

The church has learned that there is no need of antagonism between knowledge and spirituality. Knowledge and intellectual training may work evil in an undevout mind, but when consecrated to the service of Christ, learning becomes the handmaid of piety. The strength and power of the Christian Church of to-day are attributable in no small degree to the Christian colleges, that have not only encouraged mental training, but have fostered refinement and humble evangelical piety. The union of scholarly training and a holy life has raised the ministry in the public estimation so that it commands more respect and influence for good than ever before. The cause of Christ never took such hold on the popular mind, and its influence never penetrated so deeply the foundations of our social organism as it does in our day.

It is farthest from our aim to exalt and magnify the knowledge that "puffeth up," or unduly to glorify the human faculties, but we do plead that the widest opportunity be offered our youth to enlarge their knowledge, and strengthen and train their mental powers, and make the most of themselves, and that they may be consecrated to the Master's service. Men and women thus trained in our Christian colleges, and eminent alike

for learning and piety, will more and more esteem the divine revelations, and through them help to hasten the establishment of the Kingdom of righteousness on the earth.

The Students' Volunteer Movement began in 1876. It aims to awaken a deeper interest in foreign missions among college students, and to enlist their services. Within a brief period, more than 4,000 students consecrated their lives to this heroic Christian work. Already, since the movement began, 600 young men and women have entered the mission field, and thousands of others are waiting on a hesitating church to furnish the means to send them to work in foreign lands. Well did Ex-President McCosh say that the Christian Church had not witnessed such a spirit of consecration since the day of Pentecost.

The colleges have done another valuable service in awakening and strengthening in the national life a deeper sense of the value and importance of human knowledge. They are monuments of the dignity and worth of ideas, and the aspirations of the human soul.

In a new country, with its marvelous possibilities, the danger has been in having an excessive and exaggerated estimate of our national advantages, and our civilization has tended to take on a too mechanical and material character. We need to have more time to cultivate the nobler nature, and, by Christian and scholarly associations and more intimate friendships, discover and prize the fineness and sweetness of character in others, which may enrich our own life and incite us to worthy action. It is the province of higher education to help foster those conditions of mind and heart whose flexibility and natural aptitudes lead the individual "to draw ever nearer to a sense of what is indeed beautiful, graceful, and becoming." Such wisdom and goodness are of the highest practical utility in the life of a nation. The colleges have helped to offset the material tendency of our civilization by holding up high ideals and emphasizing the supremacy of the unseen mental, moral, and spiritual forces in our life. Through their leadership in the schools, and through the press, platform and pulpit, they have introduced into the fomenting mind of the republic the noblest ideals and the most generous incentives, which have, in a large measure, transformed public sentiment for the better. We have, at least, learned one great lesson in our history: that if we would have peace, contentment, happiness and prosperity, we must give the people a Christian education, and put all we can into character.

The college receives students from all ranks and conditions of society, and holds open to them its great opportunities, and worthily trains them to go forth into those professions and higher walks of life where their generous character and refreshing influences may be of larger service to the whole community. In the language of President Thwing, it may be said that "it is to the people that the college and university desire to give more than they receive from the people. It is not unjust to say that the people are debtors. The community has given to Yale, and to Princeton, and to Harvard, much, but Yale, and Princeton, and Harvard have given to the community more. For the college and the university are set to hold up the worth of things to the mind, and these things are the worthiest. In an age democratic and material, they are to represent the monarchy of the immaterial. In an age of luxuriousness, they are to declare the words of Him, homeless and pillowless, who said: 'A man's life consisteth not in the abundance of things which he hath.' They stand for the continuity of the best life, intellectual, ethical, religious, Christian. In the realm of thought, they stand for the value of ideas; in the realm of morals, for the value of ideals; in the realm of being, like the church, for the value of character."

Next to the home, the college has been the ruling spirit in private and public life. The colleges have rigorously upheld the principles of piety, justice and sacred regard for truth as the best foundation of social order. The true wealth and power of the nation are the great and good men produced by the colleges whose example and influence have been to promote intelligence and good order in society.

We look over our vast territory, with its multiplied resources and growing population, and rejoice in our material possibilities and social privileges. But what is better and grander than all these, is the fact that more than 300 Christian colleges are scattered over our land as beacon lights in our national life, building up Christian character as the best legacy for present and future generations. Some of the colleges are yet weak and struggling, but they glory in their aspirations and prospects of future grandeur. The great fabric of our national life is radiant with the golden threads of good influences emanating from these centers of superior

intelligence and instruction, where time is given for careful thought and reflection on the great problems of life.

Education by the Christian college is essential to the largest growth and progress of the state, the church, and all humanitarian movements. "The progress grows more rapid," says William T. Harris, "as the Christian spirit which leavens our civilizations sends forward, one after another, its legions into the field; for great inventions, as well as great moral reforms, proceed from Christianity."

No one can afford to be indifferent to the power and influence for good of the Christian college. These are immeasurable. The Christian Church and all the friends of human progress and welfare must, more and more, emphasize the lesson that, if we educate in our colleges the leading minds of the nation, we will be able so to control the prevailing habits and modes of thought throughout the country as to secure the permanency and glory of Christian liberty and religious institutions.

These truths may be enforced by many historic examples. The Jesuits have always been eminent for their adroit management of men. They recovered a large part of Europe to the papacy by seizing and controlling the colleges and universities as fountains of power. They had at one time under their control 600 colleges. They made it their business to educate the leading minds, and through them to guide and govern communities and nations. When only one in thirty of the inhabitants of Austria adhered to the papacy, Professor Ranke says that "the Jesuits obtained a controlling influence in the universities, and in a single generation Austria was lost to the Reformation and regained to the papal hierarchy."

In the sixteenth century, the Protestant King of Poland appointed a Jesuit minister of public instruction, who soon filled the professors' chairs with members of his own order. The "scale was soon turned, and the doctrines of the Reformation never again recovered the ascendancy."

In our own day, the influence of a college education is seen in the case of a number of young Bulgarians at Roberts College, in Constantinople. These students rekindled hope and courage in the people and revived the feeling of nationality in the hearts of the Bulgarians. This prepared the way for a general uprising in 1876, the bloody repression of which brought on the war with Russia, which led to the liberation of the province. Thus, influences descend with power from above into society. The colleges are the right arm of strength for all noble efforts for human welfare. Professor Van Holst, in his recent address, delivered at Chicago, said: "The most effectual way to lift the masses to a higher plane—materially, intellectually and morally—is to do everything favoring the climbing up of an ever-increasing minority to higher and higher intellectual and moral altitudes. Therefore, universities of the very highest order become every year more desirable—nay, necessary—for the preservation and the development of the vital forces of American democracy. Undoubtedly, to have them established is the interest of those who would frequent them, but it is still infinitely more in the interests of the American people in its entirety."

It is impossible to estimate all the good that comes to society through the influence of the college. It is quite evident that our colleges stand for the production of the highest manhood and womanhood, and their friends should marshal their forces to enhance their growth and usefulness. It is the underlying forces at work for good in our colleges that insure the integrity and safety of our social and religious organizations. Men and women who have means should regard it a privilege to lavish their gifts upon the colleges that labor for the imperishable things of life, and provide incentives for the highest Christian character and activity. He who consecrates his money to found a professorship in a Christian college erects a monument to the worth of the human soul, and perpetuates his own fame. He helps the colleges to determine, in a large measure, the character of the persons who shall fill our pulpits, teach our schools, edit our papers, write our books, and give direction to all the political and social movements. The dangers that menace our nation lie in the lack of intelligent Christian leadership. It is within the power of friends of the colleges to enroll among the college graduates a vast army of the youth of our land, whose largeness of manhood and womanhood and magnificence of character will commend themselves to the love and esteem of the lowly and suffering in every land.

Lord Macaulay once said that "the destiny of England is in the great heart of England," and we may safely say that the power for usefulness of the colleges is in the great heart of the Christian people of America, who will be more and more loyal to the sacred trust.

TRANSCRIBER'S NOTE.

The ordering of the [table](#) in Chapter II has been left as originally printed, although Dartmouth and Queen's Rutgers are not in chronological order.

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